MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

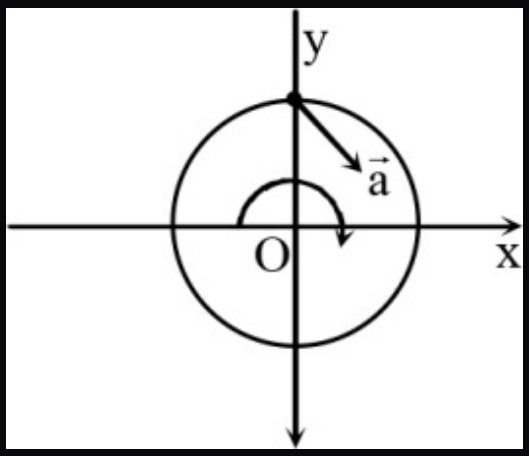
GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

A body is moving is $x-y$ plane as shown in a circular path of radius $2\ \mathrm{m}$. At a certain instant when the body is crossing the positive y -axis its acceleration is $\left(6\hat{i}-8\hat{j}\right)\ \mathrm{m\ s^{-2}}$. Then its angular acceleration and angular velocity at this instant will be-



(A) $-\ 3\hat{k}\ \mathrm{rad\ s^{-2}}$ and $-2\hat{k}\ \mathrm{rad\ s^{-1}}$ respectively

(B) $+3\hat{k}\ \mathrm{rad\ s^{-2}}$ and $+2\hat{k}\ \mathrm{rad\ s^{-1}}$ respectively

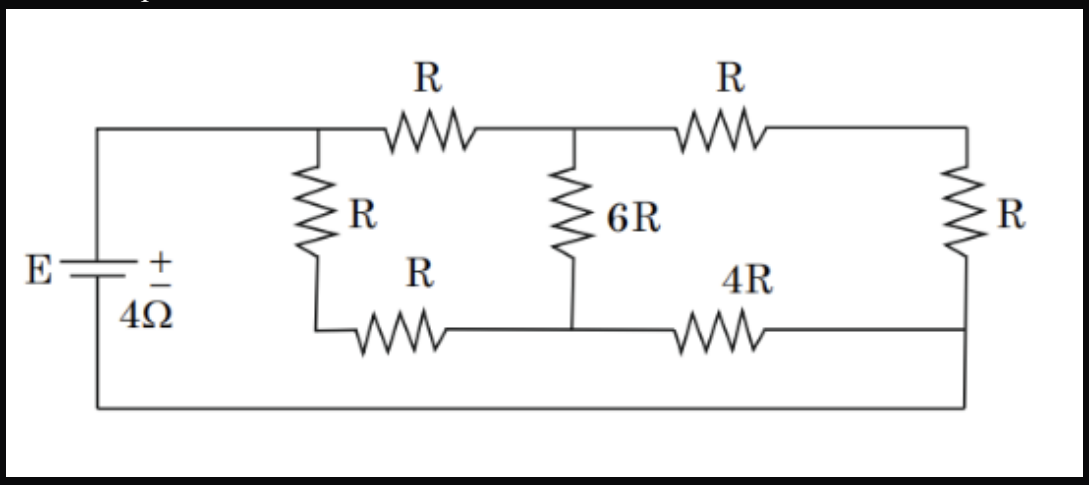
(C) $-4\hat{k}\ \mathrm{rad\ s^{-2}}$ and $-\sqrt{3}\ \hat{k}\ \mathrm{rad\ s^{-1}}$ respectively

(D) $+4\hat{k}\ \mathrm{rad\ s^{-2}}$ and $+\sqrt{3}\ \hat{k}\ \mathrm{rad\ s^{-1}}$ respectively

Q.2

A battery of internal resistance 4Ω is connected to the network of resistances as shown below. In order to deliver the

maximum power to network, the value of R in ohms should be



(A) $\frac{4}{9}$

(B) 2

(C) $\frac{8}{3}$

(D) 18

Q.3

The acceleration due to gravity 'g' and mean density of the earth ρ are related by which of the following relations? (Where, G is the gravitational constant and R is radius of earth.)

(A) $\rho = \frac{4\pi gR^2}{3G}$

(B) $\rho = \frac{4\pi gR^3}{3G}$

(C) $\rho = \frac{3g}{4\pi GR}$

(D) $\rho = \frac{3G}{4\pi GR^3}$

Q.4

Assertion- Both in fusion and fission reactions total binding energy of daughter nuclei is more than the total binding energy of parent nuclei.
Reason- In any nuclear reaction, energy is released, if total binding energy increases.

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true but Reason is not correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

Q.5

Assertion- If light continuously falls on a metal surface, then the emission of electrons will stop after some time.
Reason- We cannot extract all the electrons of a metal.

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Assertion is true but Reason is false.

(C) Assertion is false but Reason is true.

(D) Both Assertion and Reason are false.

Q.6

The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will

(A) Be independent of α and β

(B) Drop to zero when $\alpha = \beta$

(C) Go on decreasing with time

(D) Go on increasing with time

Q.7

A particle starts SHM from the mean position. Its amplitude is a and total energy E. At one instant its kinetic energy is $\frac{3E}{4}$. Its displacement at that instant is

(A) $\frac{a}{\sqrt{2}}$

(B) $\frac{a}{2}$

(C) $\frac{a}{\sqrt{\left(\frac{3}{2}\right)}}$

(D) $\frac{a}{\sqrt{3}}$

Q.8

If the intermolecular forces vanish away, the volume occupied by the molecules contained in 4.5 kg water at standard temperature and pressure will be given by

(A) 5.6 m^3

(B) 4.5 m^3

(C) 11.2 L

(D) 11.2 m^3

Q.9

Assertion: For looping a vertical loop of radius r, the minimum velocity at the lowest point should be $\sqrt{5\,gr}$
Reason: Velocity at the highest point would be zero.

(A) Assertion is True, Reason is True; Reason is a correct explanation for Assertion.

(B) Assertion is True, Reason is True; Reason is not a correct explanation for Assertion.

(C) Assertion is True, Reason is False.

(D) Assertion is False but Reason is True.

Q.10

A beaker contains water up to a height h_1 and kerosene of height h_2 above water so that the total height of (water +kerosene) is $(h_1 + h_2)$. Refractive index of water is μ_1 and that of kerosene is μ_2. The apparent shift in the position of the bottom of the beaker when viewed from above is

(A) $\left(1 - \frac{1}{\mu_1}\right)h_2 + \left(1 - \frac{1}{\mu_2}\right)h_1$

(B) $\left(1 + \frac{1}{\mu_1}\right)h_1 + \left(1 + \frac{1}{\mu_2}\right)h_2$

(C) $\left(1 - \frac{1}{\mu_1}\right)h_1 + \left(1 - \frac{1}{\mu_2}\right)h_2$

(D) $\left(1 + \frac{1}{\mu_1}\right)h_2 - \left(1 + \frac{1}{\mu_2}\right)h_1$

Q.11

Two beams, A and B of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity) a rotation of polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals:

(A) $\frac{1}{3}$

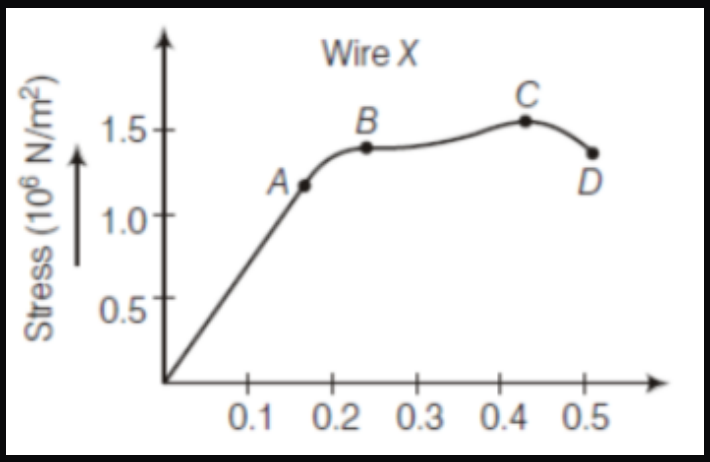
(B) 3

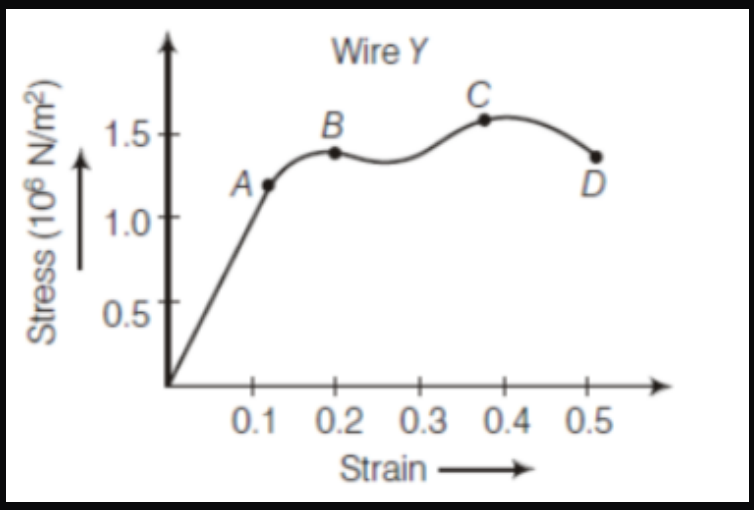
(C) $\frac{3}{2}$

(D) 1

Q.12

Consider two stress-strain graphs of two metal wires of different materials (say material X and Y)





Here, A, B, C and D represent the proportionality limit point, yield point, ultimate strength point and breaking point. Which of the following statement is correct?

(A) Wire X is more ductile than wire Y.

(B) Wire Y is more ductile than wire X.

(C) Young's modulus of wire X is more than wire Y.

(D) These wires cannot be compared as data is insufficient.

Q.13

When a loop of wire is dipped in a wetting liquid and is taken out. A liquid film is formed and a loop of light inextensible thread is gently put on the liquid film. A hole is pricked inside the loop of thread. Due to property of surface tension, free surface of the liquid tries to minimize its surface area and hence area of the hole will be maximized. In the above description, loop wire is square of side 4 units. If length of thread is 15 units, what is the final surface area for soap film on one side of wire frame? (Neglect the friction everywhere)

(A) 0.289 units^2

(B) 0.3625 units^2

(C) 1.9375 units^2

(D) 2 units^2

Q.14

If a coil of metal wire is kept stationary in a uniform magnetic field, then _____.

(A) An e.m.f. is induced in the coil.

(B) A current is induced in the coil.

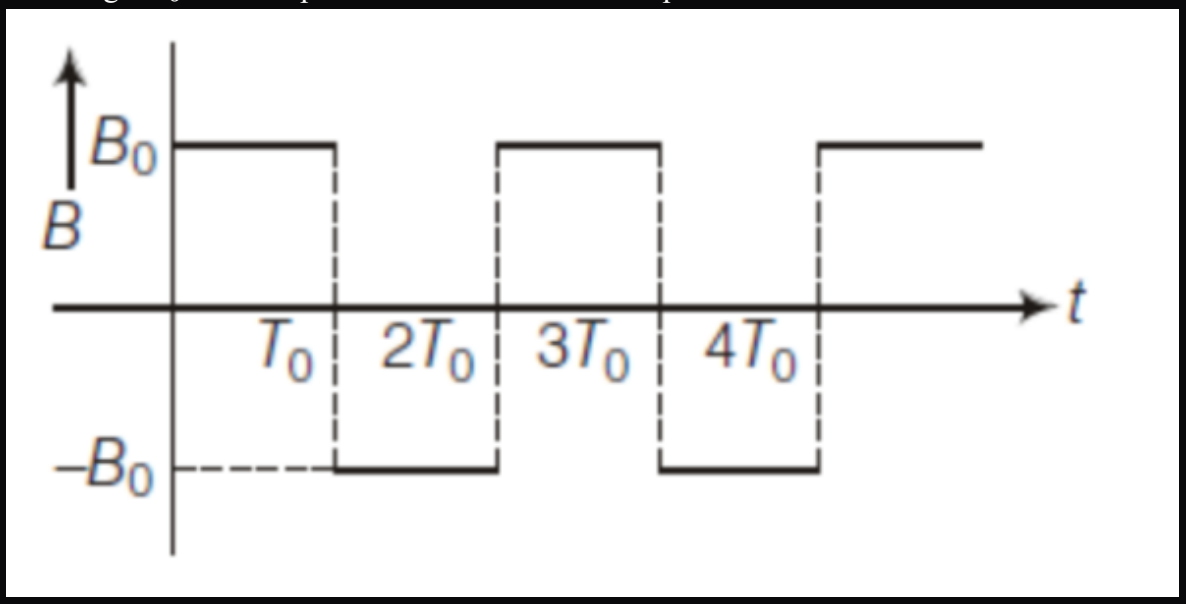
(C) Neither e.m.f. nor current is induced

(D) Both current and e.m.f. are induced

Q.15

In a region, magnetic field along X-axis changes with time according to the given graph. If time period, pitch and

radius of helix path are T_0, P_0 and R respectively, then which of the following is incorrect, if the particle is projected at an angle θ_0 with the positive X-axis in $X-Y$ plane?



(A) At $t=\frac{T_0}{2}$, coordinates of charge are $\left(\frac{P_0}{2},0,-2R_0\right)$

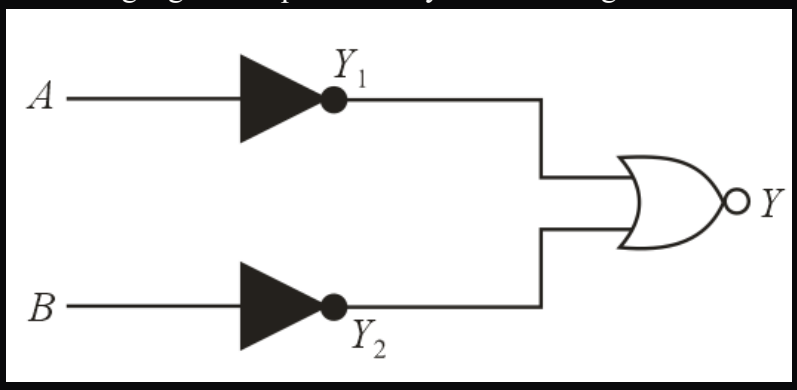
(B) At $t=\frac{3T_0}{2}$, coordinates of charge are $\left(\frac{3P_0}{2},0,2R_0\right)$

(C) Two extremes from X-axis are at a distance $2R_0$ from each other

(D) Two extremes from X-axis are at a distance $4R_0$ from each other

Q.16

Which logic gate is represented by the following combination of logic gates?



(A) NAND

(B) AND

(C) NOR

(D) OR

Q.17

Two positive ions, each carrying a charge q, are separated by a distance d. If F is the force of repulsion between the ions, the number of electrons missing from each ion will be (e being the charge of an electron)

(A) $\frac{4\pi\varepsilon_0 F d^2}{e^2}$

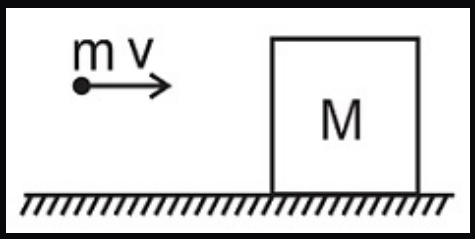
(B) $\sqrt{\frac{4\pi\varepsilon_0 F e^2}{d^2}}$

(C) $\sqrt{\frac{4\pi\varepsilon_0 F d^2}{e^2}}$

(D) $\frac{4\pi\varepsilon_0 F d^2}{q^2}$

Q.18

A bullet of mass m moving with a speed v hits a stationary block of mass M at the topmost point and gets embedded in it. If friction is sufficient to prevent slipping then the block.



(A) Must topple down

(B) Must not topple down

(C) May topple down

(D) Will remain at rest

Q.19

The charge of a parallel plate capacitor is varying as $q = q_0 \sin(2\pi f t)$. The plates are very large and close together (area $= A$, separation$= d$). Neglecting the edge effects, the displacement current through the capacitor is

(A) $\frac{d}{A\varepsilon_0}$

(B) $\frac{d}{\varepsilon_0}\sin 2\pi f t$

(C) $2\pi f q_0 \cos 2\pi f t$

(D) $\frac{2\pi f q_0}{\varepsilon_0}\cos 2\pi f t$

Q.20

The transverse displacement of a wave on string is given by $y(x,t) = \mathrm{e}^{-\left(ax^2+bt^2+2\sqrt{ab}xt\right)}$, this represents a

(A) Wave moving along $+x$-axis with speed $\sqrt{\frac{a}{b}}$

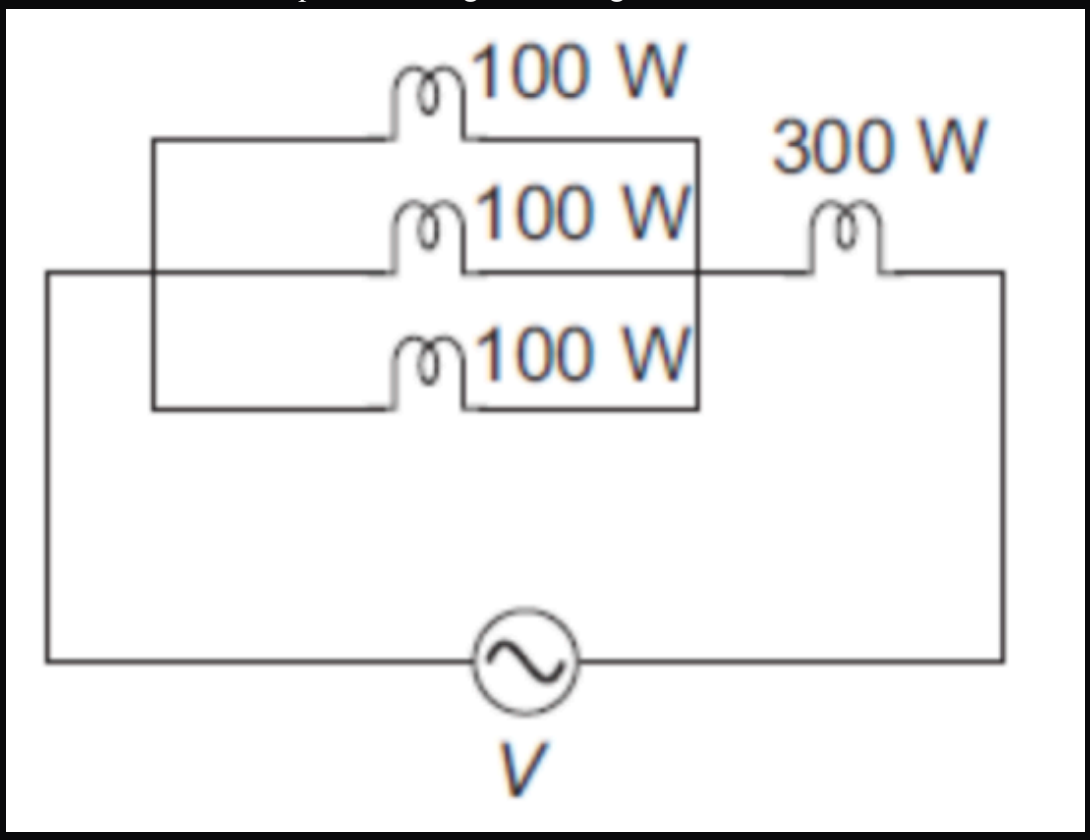
(B) Wave moving along $-x$-axis with speed $\sqrt{\frac{b}{a}}$

(C) Standing wave of frequency $\sqrt{b}$

(D) Standing wave of frequency $\frac{1}{\sqrt{b}}$

Q.21

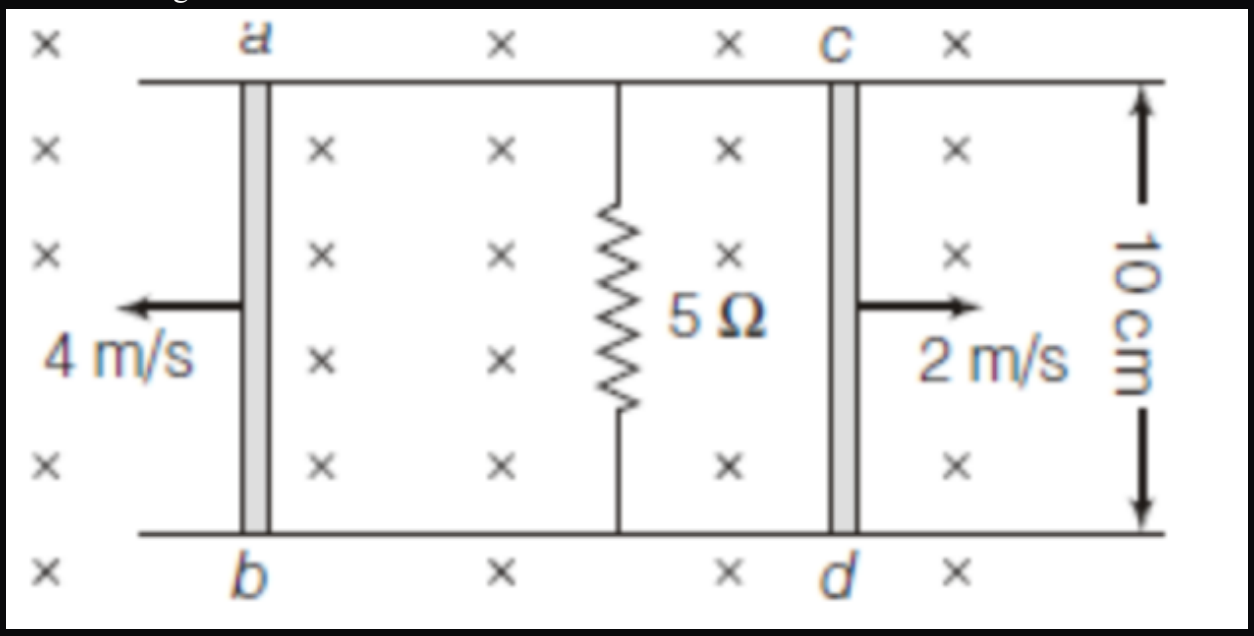
Four bulbs of different power readings are arranged in a circuit as shown below



The resultant power of combination if rated voltage is applied across the combination is .....W.

Q.22

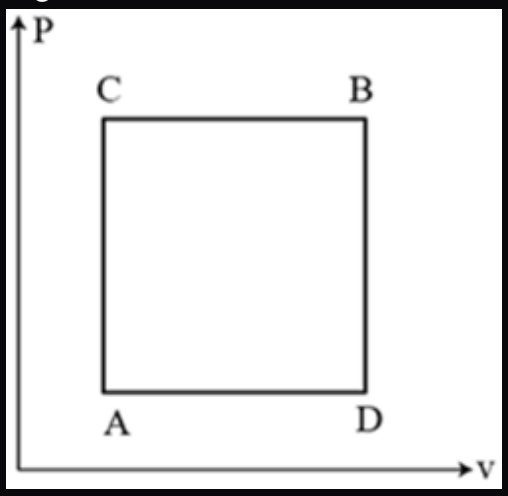
Consider the given case



Magnetic field of $1\ \mathrm{T}$ is applied to this arrangement. If connecting wires and rods have negligible resistance. Then, current through 5Ω resistor is . . . . . mA.

Q.23

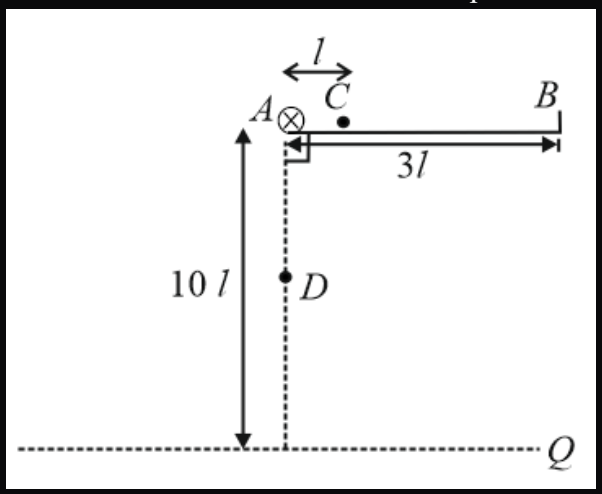
A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used $60\ \mathrm{J}$ of heat flows into the system and $30\ \mathrm{J}$ of work is done by the system. If path ADB is used work done by the system is $10\ \mathrm{J}$. The heat flow (in joule) into the system in path ADB is :

Q.24

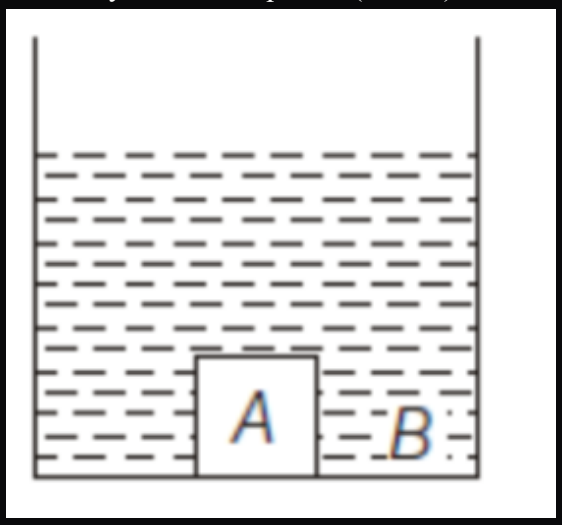
A uniform thin rod of mass 'm' and length ℓ is released from rest from horizontal position as shown. When it passes through the vertical line AD, it gets broken at point 'C' due to some reason. Find the angle in radian rotated by rod BC till. Its centre of mass passes through the horizontal line PQ from the instant of breakage. $\left(\mathrm{g}=10\ \mathrm{m/s^2}\right)$ (hinge is smooth and there is no thrust on one part of rod due to the other part at the time of breaking.)



Q.25

A parallel sided block of glass of refractive index 1.5, which is $36\ \mathrm{mm}$ thick rests on the bottom of a tank, filled with water (refractive index $=4/3$). The difference between the apparent depth of bottom at A and B, when seen from

vertically above is equal to (in mm)



Q.26

The alkene C_6H_{10} producing $OHC(CH_2)_4CHO$ on ozonolysis is:

(A) Hexene-1

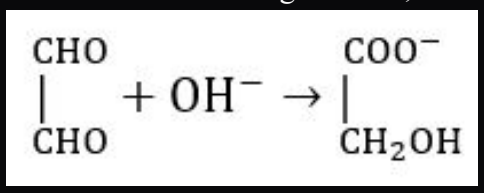
(B) Hexene-3

(C) Cyclohexene

(D) 1-Methylcyclohexene

Q.27

Consider the following reaction,



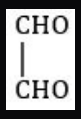
Select the incorrect statement.

(A) It is not a disproportionation reaction

(B) It is intramolecular redox reaction

(C) OH^- is a reducing as well as oxidising agent

(D)



is a reducing as well as oxidising agent

Q.28

What will be the pH of a solution by mixing $40\ mL$ of $0.1M$ HCl with $10\ mL$ of $0.45M$ $NaOH$?

(A) 12

(B) 10

(C) 8

(D) 6

Q.29

Pick the wrong statement from the following options:

(A) Deficiency of vitamin B6 (pyridoxime) results in convulsions

(B) Sources of vitamin are yeast, milk, green vegetables and cereals

(C) Deficiency of vitamin D cause Xerophthalmia

(D) Consumption of citrus fruits and green leafy vegetables in food prevents scurvy

Q.30

A Lewis acid /X/ reacts with $LiAlH_4$ in ether medium to give a highly toxic gas. This gas when heated with NH_3 gives a compound commonly known as inorganic benzene. The gas is

(A) B_2H_6

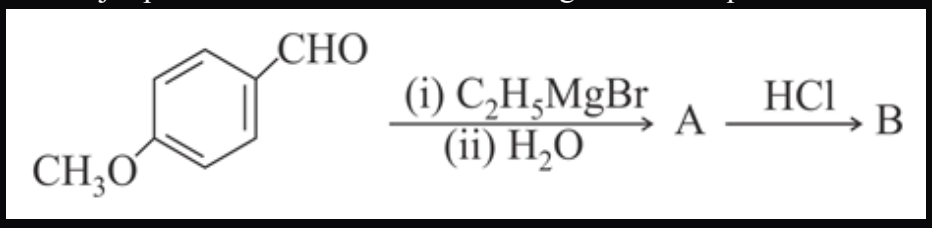
(B) $B_3N_3H_6$

(C) BF_3

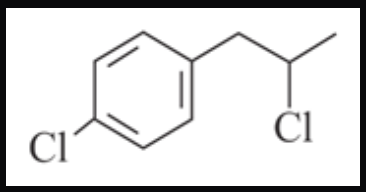
(D) B_2O_3

Q.31

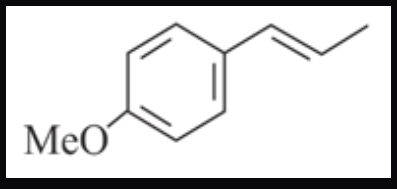
The major product B formed in the following reaction sequence is:



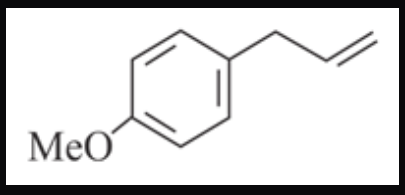
(A)



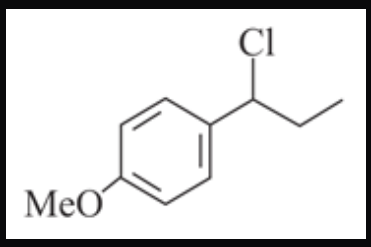
(B)



(C)



(D)



Q.32

Give the IUPAC name of $\mathrm{m}-\mathrm{ClCH_2C_6H_4CH_2C\left(CH_3\right)_3}$

(A) 1 - (3 - Chloro - 3 - methylphenyl) -2,2 - diethyl propane

(B) 2 - (3 - Chloromethyl propyl) - 2, 2 - dimethyl propane

(C) 1 - (3 - Chloromethyl phenyl) - 3, 3 - dimethyl propane

(D) 1 - Chloromethyl - $3-(2,2$ - dimethyl propyl) benzene

Q.33

Which of the following statements regarding copper salts is not true:

(A) Copper (I) disproportionates into Cu and Cu (II) in aqueous solution

(B) Copper (I) can be stabilized by the formation of insoluble complex compounds such as $\mathrm{CuCl_2^-}$ and $\mathrm{Cu\left(CN\right)_2^-}$

(C) Copper (II) oxide is red powder

(D) Hydrated $\mathrm{CuSO_4}$ is $\mathrm{[Cu(H_2O)_4]SO_4.H_2O}$

Q.34

Match the following lists:

List –I	List – II
(A) Tartaric acid	(P) Metamerism
(B) $\mathrm{CH_3\,CH=CH-CH_3}$	(Q) Cis-trans isomerism
(C) $\mathrm{CH_3-CH_2-NO_2}$	(R) Optical isomerism
(D) $\mathrm{CH_3-CH_2-O-CH_2-CH_3}$	(S) Conformational isomerism
	(T) Dynamic isomerism

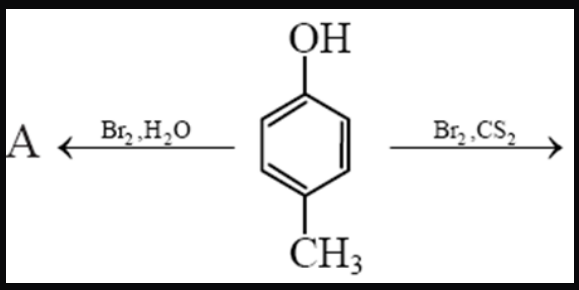
(A) $(\mathrm{A-R})(\mathrm{B-Q})\ (\mathrm{C-T})(\mathrm{D-P})$

(B) $(\mathrm{A-P})(\mathrm{B-Q})\ (\mathrm{C-T})\ (\mathrm{D-S})$

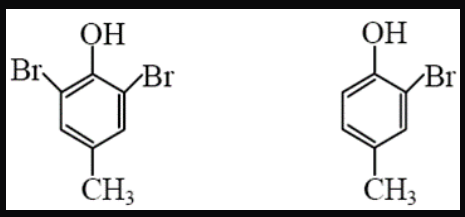
(C) $(\mathrm{A-R})(\mathrm{B-T})(\mathrm{C-Q})(\mathrm{D-P})$

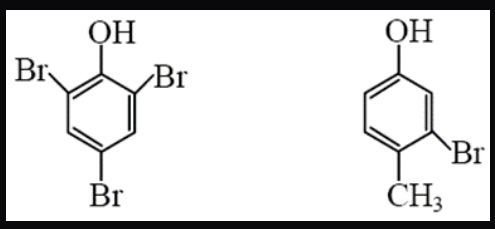
(D) $(\mathrm{A-R})(\mathrm{B-S})(\mathrm{C-T})(\mathrm{D-P})$

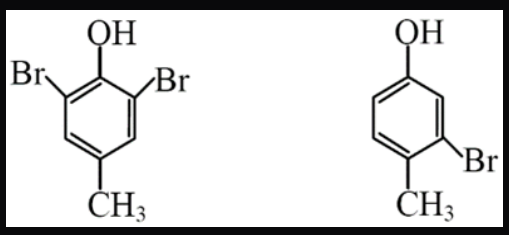
Q.35

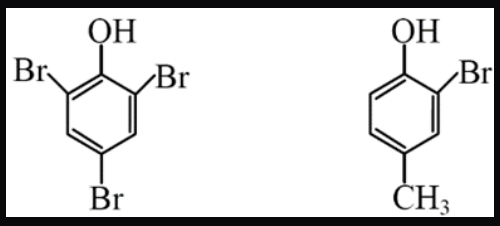


B Predominant ‘A’ and ‘B’ are respectively

(A) 

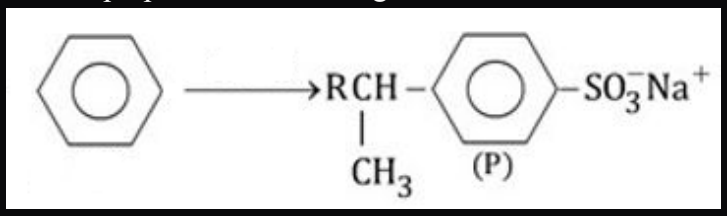
(B) 

(C) 

(D) 

Q.36

For the preparation of a detergent 'P' from benzene, The following steps are involved:



I. $\xrightarrow[\text{reaction}]{RCH=CH_2, HF, \text{Friedel-Crafts}}$

II. $\xrightarrow{H_2SO_4, SO_3}$

III. $\xrightarrow{NaOH}$

These steps should be in sequence of

(A) I, II, III

(B) II, I, III

(C) II, III, I

(D) I, III, II

Q.37

Statement 1 : Between any two given energy levels, the number of absorption transitions is always less than the number of emission transitions.

Statement 2 : Absorption transitions start from the lowest energy level and may end at any higher energy level. But emission transitions may start from any higher energy level and end at any energy level below it.

(A) Statement- 1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement -1

(B) Statement -1 is True, Statement -2 is True ; Statement- 2 is NOT a correct explanation for Statement - 1

(C) Statement - 1 is True, Statement- 2 is False

(D) Statement -1 is False, Statement -2 is True

Q.38

Amine is not formed in the reaction
I. Hydrolysis of RCN
II. Reduction of $\mathrm{RCH=NOH}$
III. Hydrolysis of RNC
IV. Hydrolysis of $\mathrm{RCONH_2}$
The correct answer is

(A) I, II and IV

(B) I and IV

(C) II and III

(D) I, II and III

Q.39

In the reaction $\mathrm{A + 3B \longrightarrow 2C}$, the rate of formation of C is

(A) the same as rate of consumption of A

(B) the same as the rate of consumption of B

(C) twice the rate of consumption of A.

(D) $3/2$ times the rate of consumption of B

Q.40

$\mathrm{K_1}$ and $\mathrm{K_2}$ are the respective equilibrium constant for the reactions
(i) $\mathrm{XeF_6(\ g) + H_2O(g) \rightleftharpoons XeOF_4(\ g) + 2HF(g); K_1}$
(ii) $\mathrm{XeO_4(\ g) + XeF_6(\ g) \rightleftharpoons XeOF_4(\ g) + XeO_3\ F_2; K_2}$
The equilibrium constant for the reaction $\mathrm{XeO_4(\ g) + 2HF(g) \rightleftharpoons XeO_3\ F_2(\ g) + H_2O(g)}$ will be

(A) $\frac{K_1}{K_2^2}$

(B) $K_1 - K_2$

(C) $\frac{K_1}{K_2}$

(D) $\frac{K_2}{K_1}$

Q.41

Fluorine has low electron affinity than chlorine because of

(A) bigger radius of fluorine, less electron density

(B) smaller radius of fluorine, high electron density

(C) smaller radius of chlorine, high electron density

(D) smaller radius of chlorine, less electron density

Q.42

A mixture of methane and ethene in a molar ratio of x : y has an average molecular mass of 20 u. The mean molar mass when they are mixed in the molar ratio of y : x will be -

(A) $20\ \mathrm{u}$

(B) $25\ \mathrm{u}$

(C) $24\ \mathrm{u}$

(D) $15\ \mathrm{u}$

Q.43

A solution is $0.10\ \mathrm{M Ba(NO_3)_2}$ and $0.10\ \mathrm{M Sr(NO_3)_2}$. If solid, $\mathrm{Na_2CrO_4}$ is added to the solution, what is $\left[\mathrm{Ba^{2+}}\right]$ when $\mathrm{SrCrO_4}$ begins to precipitate? $\left[K_{\text{sp}}\left(\mathrm{BaCrO_4}\right) = 6 \times 10^{-9},\ K_{\text{sp}}\left(\mathrm{SrCrO_4}\right) = 7 \times 10^{-6}\right]$

(A) $8.5 \times 10^{-5}\mathrm{M}$

(B) $9.5 \times 10^{-5}\mathrm{M}$

(C) $8.5 \times 10^{-9}\mathrm{M}$

(D) $8.5 \times 10^{-6}\mathrm{M}$

Q.44

Acetic acid reacts with PCl_5 to form

(A) $\mathrm{CH_2ClCOOH}$

(B) $\mathrm{CHCl_2COOH}$

(C) $\mathrm{CH_3COCl}$

(D) $\mathrm{CH_3COOCl}$

Q.45

Statement 1 : Tertiary amine is less basic than secondary amine in aqueous solution.

Statement 2 : Over crowding due to I^+ effect of three alkyl groups increases strain.

(A) Statement 1 is true, Statement 2 is true; Statement 2 is correct explanation for Statement 1.

(B) Statement 1 is true, Statement 2 is true; Statement 2 is not correct explanation for Statement 1.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is false, Statement 2 is true.

Q.46

How many of the following are lanthanides?

Uranium, praseodymium, erbium, gadolinium, curium, hafnium, osmium, iridium

Q.47

Relative decrease in vapour pressure is 0.4 for a solution containing 1 mole AB in 3 mol H_2O, therefore AB is ionised at the extent of (mark answer in percentage)

Q.48

The rate of decomposition of a substance increases 2.25 times, when the concentration of reactant is increased by 1.5 times at same T. The order of reaction is

Q.49

The maximum number of electrons that can have principal quantum number, $\mathrm{n} = 3$, and spin quantum number, $\mathrm{m_s} = -\frac{1}{2}$, is

Q.50

How many dichloro products (including stereoisomers) will be formed when $\mathrm{R}-2-$ chloropentane reacts with Cl_2 in presence of UV radiation?

Q.51

If A, B are two non-singular matrices of order 3 and I is an identity matrix of order 3 such that $AA^T = 5\mathrm{I}$ and $3A^{-1} = 2A^T - A\text{adj}(4B)$, then $|B|^2$ is equal to (where A^T and $\text{adj}(A)$ denote transpose and adjoint matrices of the matrix A respectively)

(A) $\frac{7^3}{5^3 \cdot 4^6}$

(B) $\frac{7^3}{5^3 \cdot 4^3}$

(C) $\frac{7^4}{5^3.2^{12}}$

(D) $\frac{5^6}{7^3.2^{10}}$

Q.52

$\int_{\log \sqrt{\pi/2}}^{\log \sqrt{\pi}} e^{2x} \sec^2\left(\frac{1}{3}e^{2x}\right) dx$ is equal to

(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) $\frac{3\sqrt{3}}{2}$

(D) $\frac{1}{2\sqrt{3}}$

Q.53

The value of $\lim_{n\to\infty}\left(\frac{[1^2x+1^2]+[2^2x+2^2]+...+[n^2x+n^2]}{n^3}\right)$ is: (where $[\cdot]$ denotes the greatest integer function.)

(A) $\frac{x}{3}$

(B) $x+\frac{1}{3}$

(C) $\frac{x}{3}+\frac{1}{3}$

(D) None of these

Q.54

The coefficient of x^4 in the expansion $(1-x)^8(1+x)^{12}$ is equal to

(A) 1

(B) -48

(C) 28

(D) -19

Q.55

The possible value of the ordered triplet (a,b,c) such that the function $f(x)=x^3+ax^2+bx+c$ is a monotonic function is

(A) $(2,1,3)$

(B) $(1,-1,3)$

(C) $(2,2,4)$

(D) $(2,-2,1)$

Q.56

Among the 8! permutations of the digits $1, 2, 3, \ldots\ldots\ldots$, consider those arrangements which have the following property. If we take any five consecutive positions, the product of the digits in these positions is divisible by 5. The number of such arrangement is equal to

(A) $7!$

(B) $2.(7!)$

(C) ${}^{7}\mathrm{C}_{4}$

(D) None of these

Q.57

Which of the following is an even function?

(A) $f(x) = \log\left(\frac{1-x}{1+x}\right)$

(B) $f(x) = \log\left(x + \sqrt{1+x^2}\right)$

(C) $\mathrm{f(x)} = \frac{\mathrm{x}}{\mathrm{e^x}-1} + \frac{\mathrm{x}}{2} + 1$

(D) $f(x) = e^{2x} + \sin x$

Q.58

In a harmonic progression $t_1, t_2, t_3, \ldots\ldots\ldots\ldots$, it is given that $t_5 = 20$ and $t_6 = 50$. If S_n denotes the sum of first n terms of this, then the value of n for which S_n is maximum is

(A) 6

(B) 7

(C) 9

(D) 10

Q.59

Consider the $A(-1, 2, 1)$ and $B(4, 3, 5)$ are two given points. Then, the projection of AB on a line which makes angle 120°, and 135° with Y and Z-axis respectively and an acute angle with X-axis is

(A) $\left(2 - 2\sqrt{2}\right)$ units

(B) $\left(2 + 2\sqrt{2}\right)$ units

(C) $\left(2 + 3\sqrt{2}\right)$ units

(D) $\left(2 + \sqrt{2}\right)$ units

Q.60

For a parabola passing through $(1, 2)$, $(2, 1)$, $(3, 4)$ and $(4, 3)$, the equation of axis of the parabola is-

(A) $x - y + 3 = 0$

(B) $x - y = 0$

(C) $x + y - 1 = 0$

(D) line $x + y = 0$

Q.61

If $\vec{r} \cdot \hat{i} = 2\vec{r} \cdot \hat{j} = 4\vec{r} \cdot \hat{k}$ and $\left|\vec{r}\right| = \sqrt{84}$, then the value of $\vec{r} \cdot \left(2\hat{i} - 3\hat{j} + \hat{k}\right)$ may be

(A) 0

(B) 2

(C) 4

(D) 6

Q.62

There are two bags, bag 1 contains 3 white and 2 red balls and bag 2 contains 4 white and 5 red balls. A ball is drawn randomly from bag 1 and put in bag 2. Now 2 balls are drawn from bag 2 and found to be red. Then the probability that white ball was drawn from bag 1 to bag 2 is

(A) $< \frac{1}{2}$

(B) $< \frac{1}{4}$

(C) $> \frac{3}{4}$

(D) $> \frac{1}{4}$

Q.63

Let $f(x) = \text{Maximum } \left\{x^2, (1-x)^2,\ 2x(1-x)\right\}$, where $0 \leq x \leq 1$. Determine the area of the region bounded by the curves $y = f(x)$, x-axis, $x = 0$ & $x = 1$

(A) $\frac{17}{27}$

(B) $\frac{15}{25}$

(C) $\frac{13}{23}$

(D) $\frac{14}{29}$

Q.64

If $8f(x) + 6f\left(\frac{1}{x}\right) = x + 5$ and $y = x^2\ f(x),$ then $\frac{dy}{dx}$ at $x = -1$ is equal to

(A) 0

(B) $\frac{1}{14}$

(C) $-\frac{1}{14}$

(D) 1

Q.65

The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3\mathrm{a}$ is

(A) $\mathrm{x}^2+\mathrm{y}^2=9\mathrm{a}^2$

(B) $x^2+y^2=16a^2$

(C) $\mathrm{x}^2+\mathrm{y}^2=4\mathrm{a}^2$

(D) $\mathrm{x}^2+\mathrm{y}^2=\mathrm{a}^2$

Q.66

Suppose $\alpha, \beta, \mathrm{y} \in \mathrm{R}$ are such that $\sin\alpha, \sin\beta, \sin\gamma \neq 0$ and $\Delta = \begin{vmatrix} \sin^2\alpha & \sin\alpha\cos\alpha & \cos^2\alpha \\ \sin^2\beta & \sin\beta\cos\beta & \cos^2\beta \\ \sin^2\gamma & \sin\gamma\cos\gamma & \cos^2\gamma \end{vmatrix}$ then

Δ cannot exceed

(A) 1

(B) 0

(C) $-\frac{1}{2}$

(D) None of these

Q.67

If R is a relation on the set N, defined by $\{(x, y) : 2x - y = 10\}$, then R is

(A) reflexive

(B) symmetric

(C) transitive

(D) none of these

Q.68

Find the ordered triplet $(A,\ B,\ \lambda)$, If $\int \frac{2\cos x - \sin x + \lambda}{\cos x + \sin x - 2}\,dx = A\log_e(|\cos x + \sin x - 2|) + Bx + C$.

(A) $\left(\frac{1}{2}, \frac{3}{2}, -1\right)$

(B) $\left(\frac{3}{2}, \frac{1}{2}, -1\right)$

(C) $\left(\frac{1}{2}, -1, -\frac{3}{2}\right)$

(D) $\left(\frac{3}{2}, -1, -\frac{1}{2}\right)$

Q.69

A stick of length a cm rests against a vertical wall and the horizontal floor. If the foot of the stick slides with a constant

velocity of $b\frac{\text{cm}}{\text{s}}$, then the magnitude of the velocity of the middle point of the stick when it is equally inclined with the floor and the wall, is

(A) $\frac{b}{\sqrt{2}}\frac{\text{cm}}{\text{s}}$

(B) $\frac{b}{2}\frac{\text{cm}}{\text{s}}$

(C) $\frac{ab}{2}\frac{\text{cm}}{\text{s}}$

(D) $\frac{a}{\sqrt{2}}\frac{\text{cm}}{\text{s}}$

Q.70

The solution of the differential equation $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$ is

(A) $\tan y \tan x = c$

(B) $\frac{\tan y}{\tan x} = c$

(C) $\frac{\tan^2 x}{\tan y} = c$

(D) None of these

Q.71

If the number of ways of selecting two distinct numbers from the set {1, 2, ... , 12} whose sum is divisible by 3 is equal to k, then k - 16 is equal to .....

Q.72

If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the vertices of an equilateral triangle whose orthocenter is at the origin, then $\vec{a}+\vec{b}+\vec{c}=$

Q.73

A function $f : A \to B$ such that $n(A) = 5$ and $n(B) = 4$ then if total number of onto function is λ then sum of digits of λ equals to

Q.74

Suppose z_1, z_2, z_3 are vertices of an equilateral triangle in an argand plane inscribed in the circle $|z| = 2$. If $z_1 = 1 - i\sqrt{3},\ z_2 = a + ib,\ z_3 = c + id$, where $a, b, c, d \in R$ and $a > c$, then $|3a + d - c|$ equals to [Note: $i = \sqrt{-1}$]

Q.75

Let $x + \frac{1}{x} = 2$, $y + \frac{1}{y} = -2$ and $\sin^{-1} x + \cos^{-1} y = m\pi$, then the value of 2m is

Answers & Solutions

Q.1 Answer:

—

Solution:

Given that, the radius of the circular path is, $r = 2\ \mathrm{m}$ **and the net acceleration of the particle is,**

$\vec{a} = \left(6\hat{i} - 8\hat{j}\right)\ \mathrm{m\ s^{-2}} \quad \cdots(1)$

Compare the above equation with

$a = \left(a_t\hat{i} - a_r\hat{j}\right)\ \mathrm{m\ s^{-2}} \quad \cdots(2)$

By comparing equation (1) **&** (2)**, we can write,**

Radial acceleration as, $a_r = -8\hat{j}\ \mathrm{m\ s^{-2}}$

Tangential acceleration as, $a_t = 6\hat{i}\ \mathrm{m\ s^{-2}}$

Relation between the tangential acceleration and angular acceleration is,

$|a_t| = 6 = r\alpha \ \Rightarrow\ 2\alpha = 6$

$\alpha = 3\ \mathrm{rad\ s^{-2}} \ \Rightarrow\ \alpha = -3\hat{k}\ \mathrm{rad\ s^{-2}}$

Relation between the radial acceleration and angular velocity is,

$|a_r| = r\omega^2 \ \Rightarrow\ \omega = \sqrt{\frac{|a_r|}{r}} \ \Rightarrow\ \omega = \sqrt{\frac{8}{2}}$

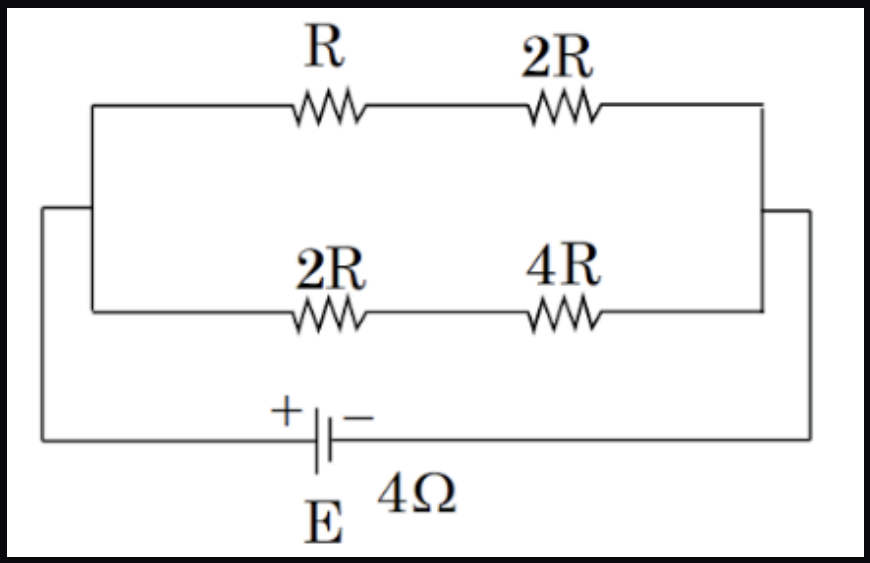
$\omega = 2\ \mathrm{rad\ s^{-1}} \ \Rightarrow\ \omega = -2\hat{k}\ \mathrm{rad\ s^{-1}}$

Q.2 Answer:

2

Solution:

Given circuit is a blanced wheat stone bridge, so no current flows through $6\mathrm{R}$ **registor and equivalent circuit is**



So, external resistance

$= \mathrm{Reg} = \frac{6\mathrm{R}\times 3\mathrm{R}}{6\mathrm{R}+3\mathrm{R}} = 2\mathrm{R}$

For maximum power output, when

$\mathrm{R_{eq}} = 4\Omega$

$\Rightarrow 2\mathrm{R} = 4 \Rightarrow \mathrm{R} = 2\Omega$

Q.3 Answer:

$\rho = \frac{3g}{4\pi GR}$

Solution:

$g = \frac{GM}{R^2}$

$g = \frac{G\frac{4}{3}\pi R^3\rho}{R^2}$

$g = \frac{4}{3}\pi GR\rho \Rightarrow \rho = \frac{3g}{4\pi GR}.$

Q.4 Answer:

Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Solution:

Energy released during a nuclear process is given as:

$Q = \Delta mc^2 = (B.E.)_{\text{daughter}} - (B.E.)_{\text{parent}}$

So, for Q to be positive,

$(B.E.)_{\text{daughter}} > (B.E.)_{\text{parent}}$

$\therefore$ **The energy will be released in nuclear reaction.**

Q.5 Answer:

Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Solution:

We can not extract all electrons from metal, as we remove electrons the metal will get more and more positively charged.

This is so because the number of proton will be more than no of electron.

Now, Attraction will be greater. So, It takes too much energy to extract the electrons any more.

Q.6 Answer:

Go on increasing with time

Solution:

Given $x = ae^{-\alpha t} + be^{\beta t}$

Velocity, $v = \frac{dx}{dt} = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$

$\text{or, } v = -\frac{a\alpha}{e^{\alpha t}} + b\beta e^{\beta t}$

Now, as time increases;

term $1 = \frac{-a\alpha}{e^{\alpha t}}$ **will decrease**

and term $2 = b\beta e^{\beta t}$ **will increase**

$\therefore$ ***v* will increase as time increases.**

Q.7 Answer:

$\frac{a}{2}$

Solution:

$$K = \frac{1}{2}m\omega^2\left(a^2 - y^2\right)$$
$$\frac{3}{4}E = \frac{1}{2}m\omega^2\left(a^2 - y^2\right)$$
$$\frac{3}{4}\left(\frac{1}{2}m\omega^2 a^2\right) = \frac{1}{2}m\omega^2\left(a^2 - y^2\right)$$
$$y^2 = a^2 - \frac{3}{4}a^2 = \frac{a^2}{4}$$
$$\Rightarrow y = \frac{a}{2}$$

Q.8 Answer:

5.6 m^3

Solution:

$n = \frac{\text{Mass of water}}{\text{Molecular wt. of water}} = \frac{4.5\text{ kg}}{18\times10^{-3}\text{kg}} = 250$

T = 273 K and $P = 10^5$ N/m^2 (STP)

From $PV = nRT$,

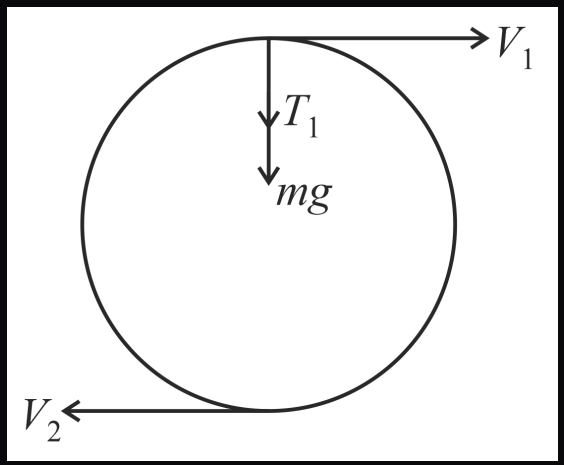
$\Rightarrow \quad V = \frac{nRT}{P} = \frac{250\times8.3\times273}{10^5} = 5.66\text{ m}^3$

Q.9 Answer:

Assertion is True, Reason is False.

Solution:

To just complete the loop, speed v_1 at the highest point must be minimum.



Applying Newton's 2nd law at the highest point.

$mg + T_1 = \frac{mv_1^2}{R}$.

For minimum v_1, tension must be minimum which can be zero, i.e. $T_1 = 0$.

$\therefore\ v_1 = \sqrt{Rg} \quad \ldots.(1)$.

Now from conservation of mechanical energy,

Loss in gravitational PE =Gain in KE

$\Rightarrow mg(2R) = \frac{1}{2}\text{mv}_2^2 - \frac{1}{2}\text{mv}_1^2$

$\Rightarrow v_2^2 = v_1^2 + 4gR$

$\Rightarrow v_2 = \sqrt{5gR} \quad [\text{Using } (1)]$.

Hence, assertion is true, reason is false.

Q.10 Answer:

$\left(1-\frac{1}{\mu_1}\right)h_1+\left(1-\frac{1}{\mu_2}\right)h_2$

Solution:

Apparent shift due to a transparent material of thickness h is give by,

$\Delta h=\left(1-\frac{1}{\mu}\right)h$

$\therefore$ Apparent shift produced by water

$\triangle h_1=\left(1-\frac{1}{\mu_1}\right)h_1$

And apparent shift produced by kerosene

$\triangle h_2=\left(1-\frac{1}{\mu_2}\right)h_2$

$\triangle h=\ \triangle h_1\ +\triangle h_2=\left(1-\frac{1}{\mu_1}\right)h_1+\left(1-\frac{1}{\mu_2}\right)h_2$

Q.11 Answer:

$\frac{1}{3}$

Solution:

By law of Malus i.e. $I=I_0\cos^2\theta$

Now $I'_A=I_A\cos^2 30^\circ$

$I'_B=I_B\cos^2 60^\circ$

As $I'_A=I'_B$

$I_A\cos^2 30^\circ=I_B\cos^2 60^\circ$

$I_A\frac{3}{4}=I_B\frac{1}{4}\quad\Rightarrow\quad\frac{I_A}{I_B}=\frac{1}{3}$

Q.12 Answer:

Wire Y is more ductile than wire X.

Solution:

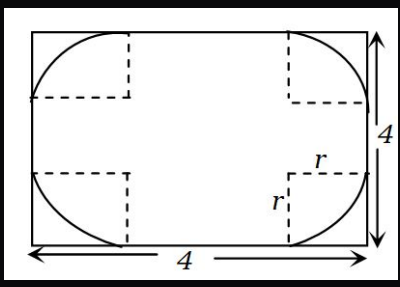
A metal is more ductile (and thus less brittle) when point of ultimate strength (i.e., C) and breaking point (i.e., D) are far apart.

So, option (b) is correct. Clearly, Young's modulus $\left(=\frac{\text{stress}}{\text{strain}}\right)$ of wire Y is more than wire X

Q.13 Answer:

0.289 units^2

Solution:



Given, square wire have side 4 unit, length of loop thread 15 unit.

From figure,

$15 = 16 - (8 - 2\pi)r$

Then, from above equation

$r = \frac{1}{8-2\pi} \ \ldots(1)$

Area of soap film will be

$= (2r)(2r) - \pi r^2 = (4 - \pi)r^2 \ \ldots(2)$

From above two equations

$A = (4 - \pi) \times \left(\frac{1}{8-2\pi}\right)^2 A = \frac{1}{4}\left(\frac{1}{4-\pi}\right)$

Then, from above result

$A = \frac{1}{4}\left(\frac{1}{4-\pi}\right) A = 0 \cdot 25\left(\frac{1}{4-3 \cdot 14}\right) = 0 \cdot 289 \ unit^2$

Q.14 Answer:

Neither e.m.f. nor current is induced

Solution:

If the coil of metal wire is kept stationary in the uniform magnetic field, there will be no change in the magnetic flux linked with the coil. Therefore, there will be no induced emf in coil. Since, there is no induced emf, there will be no induced current also in the coil.

Q.15 Answer:

Two extremes from X-axis are at a distance $2R_0$ from each other

Solution:

At $t = \frac{T_0}{2}, x = \frac{P_0}{2}$
$\theta = \pi, y = 0$ **and** $z = -2R_0$
At $t = \frac{3T_0}{2}, x = \frac{3P_0}{2}$
$\theta = \pi, y = 0$ **and** $z = 2R_0$
Maximum distance $2R_0$**.**

Q.16 Answer:

AND

Solution:

$Y_1 = \overline{A},\ Y_2 = \overline{B},\ Y = \overline{Y_1 + Y_2} = \overline{\overline{A} + \overline{B}} = A \cdot B \Rightarrow$**AND gate**

Q.17 Answer:

$\sqrt{\frac{4\pi\varepsilon_0 F d^2}{e^2}}$

Solution:

Let n **be the number of electrons missing**

$$F = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q^2}{d^2} \Rightarrow q = \sqrt{4\pi\varepsilon_0 d^2 F} = ne$$

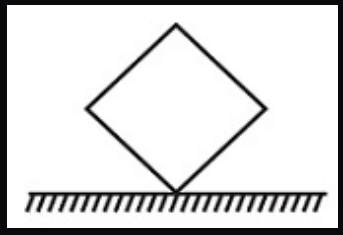
$$\therefore \quad n = \sqrt{\frac{4\pi\varepsilon_0 F d^2}{e^2}}$$

Q.18 Answer:

May topple down

Solution:

If the K.E. of the system just after the collision is sufficient to rise the block as shown, the block will topple, otherwise it will not topple down.



Q.19 Answer:

$2\pi f q_0 \cos 2\pi f t$

Solution:

Displacement current is given as rate of flow of charge during charging or discharging of the capacitor. It is given by

$i_d = \frac{dq}{dt}$

Given here $q = q_0 \sin(2\pi f t)$

Putting the value of charge in displacement current equation,

$i_d = \frac{d}{dt}(q_0 \sin 2\pi f t) = q_0 2\pi f \cos 2\pi f t$

Q.20 Answer:

Wave moving along $-x$-axis with speed $\sqrt{\frac{b}{a}}$

Solution:

General equation of travelling wave is

$y = f(ax \pm bt) = f(x \pm vt)$

$y = \mathrm{e}^{-\left(ax^2+bt^2+2\sqrt{ab}xt\right)} = \mathrm{e}^{-(\sqrt{a}x+\sqrt{b}t)^2}$

So, the wave will be travelling along $-x$**-axis with speed** $\sqrt{\frac{b}{a}}$**.**

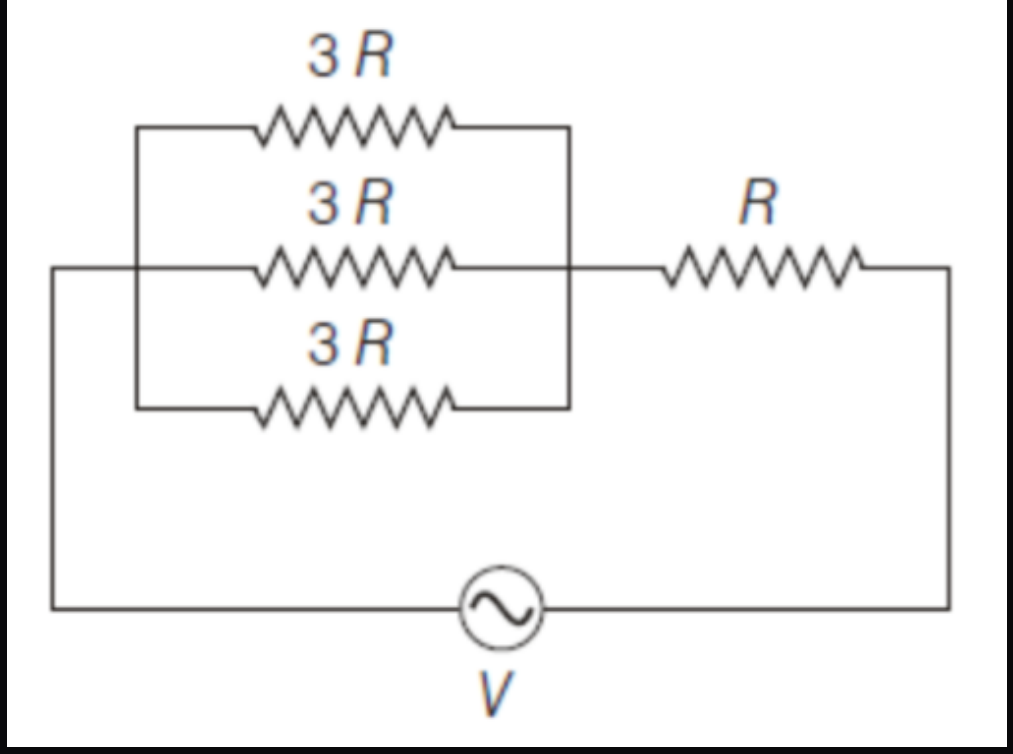
Q.21 Answer:

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Solution:

Let resistance of $300\ \mathrm{W}$ **bulb is** R**. Hence,** $\frac{V^2}{R} = 300\ \mathrm{W}$.

$\therefore$ **Resistance of** $100\ \mathrm{W}$ **bulb should be** $3R$



So, we have

$\therefore$ Equivalent resistance is $R_{\text{eq}} = 2R$

Hence, power $_{eq} = \frac{V^2}{(2R)}$

$= \frac{1}{2} \times \frac{V^2}{R} = \frac{300}{2} = 150 \text{ W}$

Q.22 Answer:

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Solution:

Potential difference induced in both the rods would have different polarity.

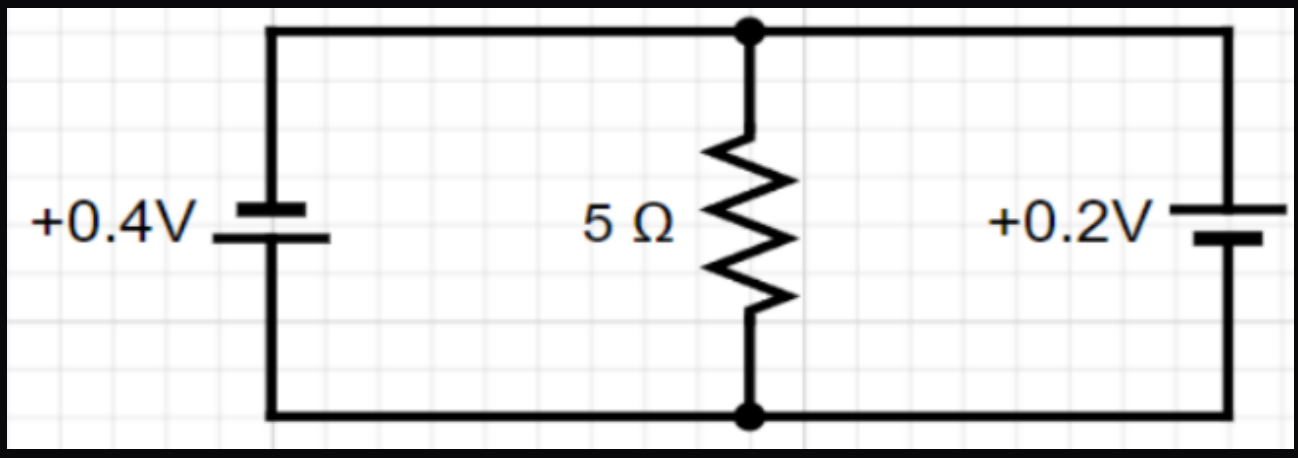
PD (potential difference) in

$ab = B/v = 1 \times 0.1 \text{ m} \times 4 \text{ m/s} = 0.4 \text{ V}$

PD (potential difference) in

$cd = B/v = 1 \times 0.1 \text{ m} \times 2 \text{ m/s} = 0.2 \text{ V}$

So, the arrangement would act like



$\therefore$ Current

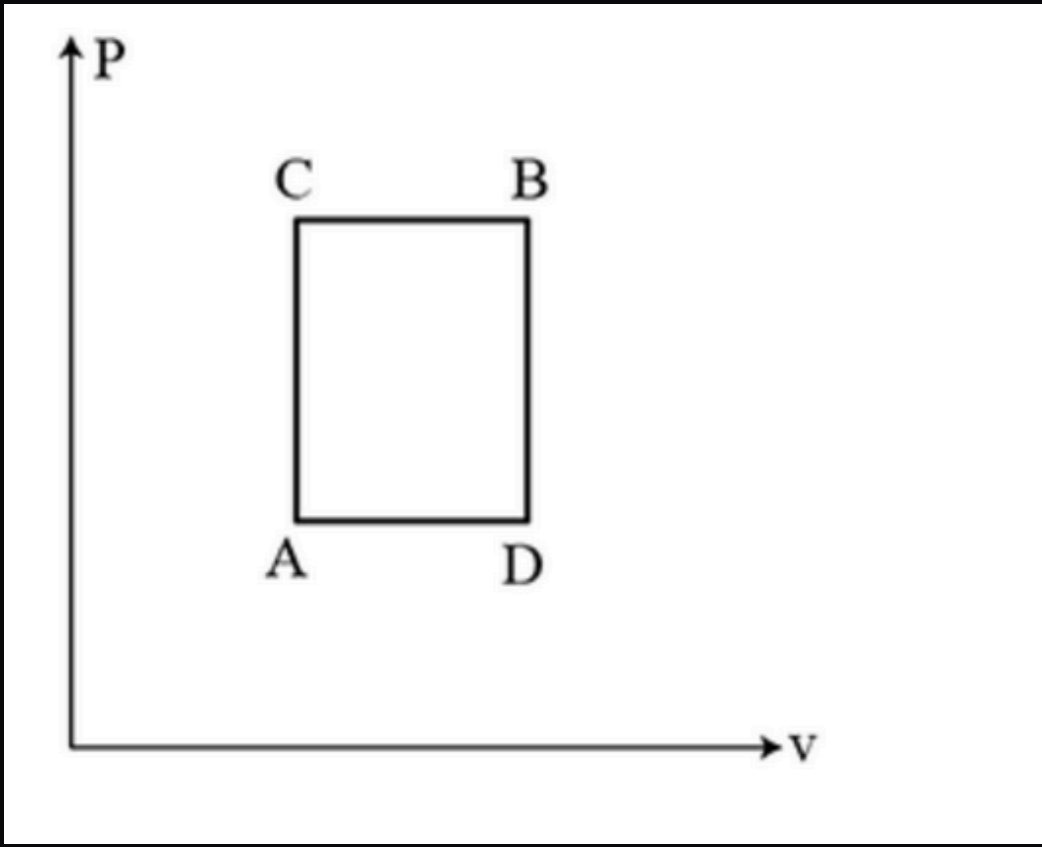
$I = \frac{0.4-0.2}{5} = \frac{0.2}{5} = 0.04 \text{ A}$

$= 40 \text{ mA}$

Q.23 Answer:

—

Solution:



Δ **Uremains same for both paths** ACB **and** ADB

$$\Delta Q_{ACB} = \Delta W_{ACB} + \Delta U_{ACB}$$

$$\Rightarrow 60\ J = 30\ J + \Delta U_{ACB}$$

$$\Rightarrow U_{ACB} = 30\ J$$

$$\therefore \Delta U_{ADB} = \Delta U_{ACB} = 30\ J$$

$$\Delta Q_{ADB} = \Delta U_{ADB} + \Delta W_{ADB}$$

$$= 10\ J + 30\ J = 40\ J$$

Q.24 Answer:

—

Solution:

By energy conservation

$$mg.\frac{3\ell}{2} = \frac{1}{2}.\frac{m(3\ell)^2}{3}.\omega^2$$

$$\omega = \sqrt{\frac{g}{\ell}}$$

And velocity of centre of mass of rod BC $V_{cm} = \sqrt{\frac{g}{\ell}}.2\ell$

$$V_{cm} = 2\sqrt{g\ell}$$

If time taken by centre of mass of rod BC from breaking position to line PQ is t.

$$8\ell = \frac{1}{2} \times g \times t^2$$

$t = 4\sqrt{\frac{\ell}{g}}$

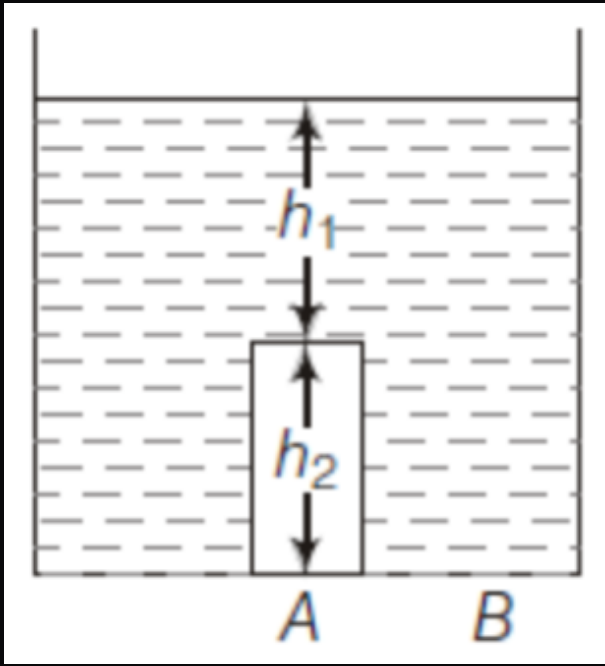
$\theta = \omega . t = 4$ radian

Q.25 Answer:

—

Solution:

Apparent depth of A,



$d_1 = \frac{h_1}{4/3} + \frac{h_2}{3/2} = \frac{3}{4}h_1 + \frac{2}{3}h_2 \quad \ldots$ **(i)**

Apparent depth of B

$d_2 = \frac{(h_1+h_2)}{4/3} = \frac{3}{4}(h_1 + h_2) \quad \ldots$ **(ii)**

From Eqs. (i) and (ii), we get

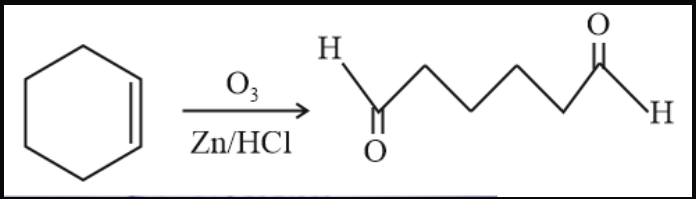
$\therefore d_2 - d_1 = \left(\frac{3}{4} - \frac{2}{3}\right) h_2 = 3$ mm

Q.26 Answer:

Cyclohexene

Solution:

Since the alkene (C_6H_{10}) on ozonolysis gives a dialdehyde six carbon containing compound therefore, the alkene must be cyclohexene.



Q.27 Answer:

OH^- is a reducing as well as oxidising agent

Solution:

One CHO is oxidised to COO^- and one CHO is reduced to CH_2OH .

Thus, it is not a disproportionation reaction. It is intramolecular redox reaction. Thus, option A and B are true and



is reducing as well as oxidising agent.

Thus, option D is also true and option C is incorrect.

Q.28 Answer:

12

Solution:

$40\ \text{mL of}\ 0.1\text{MHCl} = \frac{0.1}{1000} \times 40$

$= 0.004\ \text{mole}$

$10\ \text{mL of}\ 0.45\text{MNaOH} = \frac{0.45}{1000} \times 10 = 0.0045\ \text{mole}$

$\therefore \text{NaOH left unused} = 0.0005\ \text{mole}$

$\text{Volume of solution} = 50\ \text{mL}$

$\therefore \text{Molarity of NaOH} = \frac{0.0005}{50} \times 1000$

$= 0.01\text{M}$

$\text{Or}\ \left[\text{OH}^-\right] = 10^{-2}\ \text{or}\ \left[\text{H}^+\right] = 10^{-12} \quad \text{pH} = 12$

Q.29 Answer:

Deficiency of vitamin D cause Xerophthalmia

Solution:

Xerophthalmia is medical condition in which the eye fails to produce tear. It may be caused by vitamin A deficiency.

Vitamin D deficiency causes rickets.

Q.30 Answer:

B_2H_6

Solution:

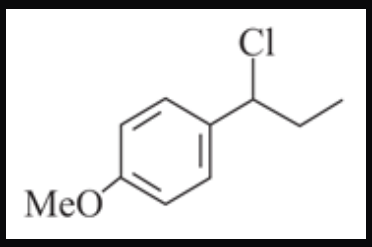
Lewis acid is boron trifluoride BF_3.

Reaction sequence of the above statement is as follow:

$4\,BF_3 + 3\,LiAlH_4 \rightarrow 2B_2H_6 + 3LiF + 3\,AlF_3$

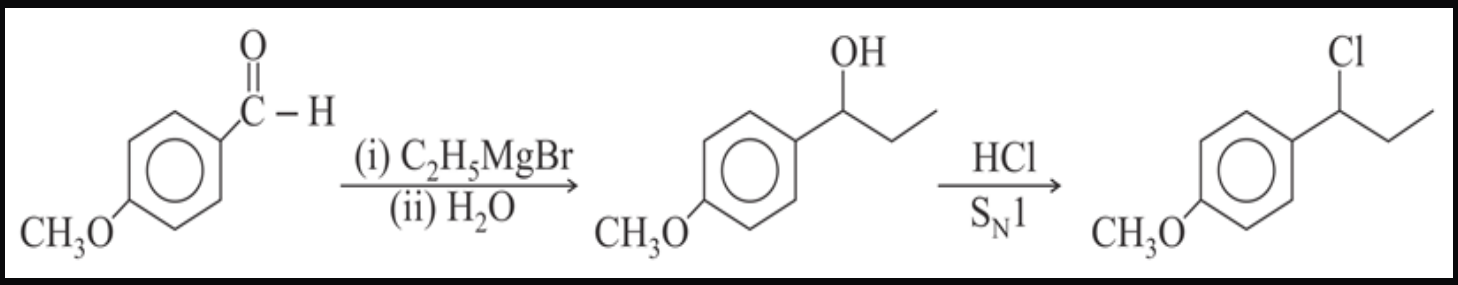
$3B_2H_6 + 6\,NH_3 \rightarrow 2B_3N_3H_6 + 12H_2$

Q.31 Answer:



Solution:

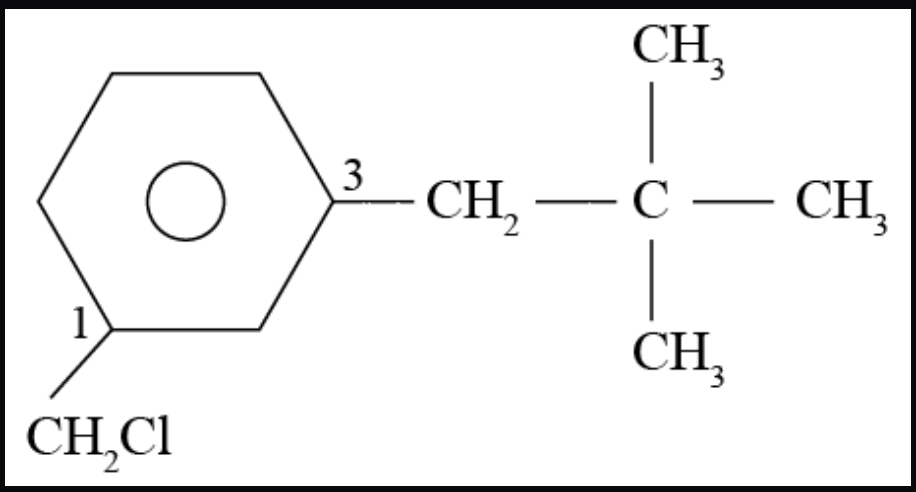
Initially aldehyde reacts with Grignard reagent to form a secondary alcohol. Which undergo nucleophilic substitution reaction.



Q.32 Answer:

1 - Chloromethyl - $3-(2,2$ - dimethyl propyl) benzene

Solution:



Q.33 Answer:

Copper (II) oxide is red powder

Solution:

Copper (II) oxide is black powder

Q.34 Answer:

$(A-R)(B-Q)\ (C-T)(D-P)$

Solution:

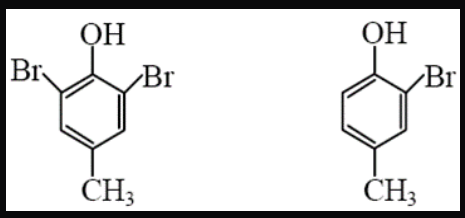
List –I	List – II
(A) **Tartaric acid**	(R) **Optical isomerism**

(B) $CH_3CH = CH - CH_3$ (Q) **Cis-trans isomerism**

(C) $CH_3 - CH_2 - NO_2$ **(T) Dynamic isomerism**

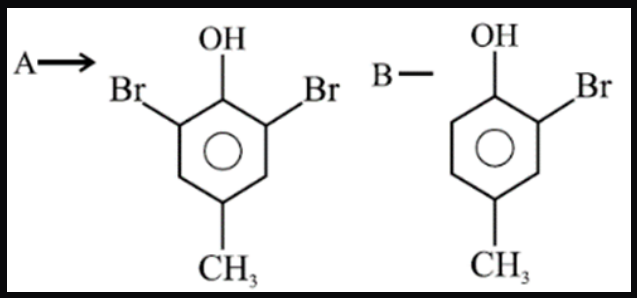
(D) $CH_3 - CH_2 - O - CH_2 - CH_3$ **(P) Metamerism**

Q.35 Answer:



Solution:

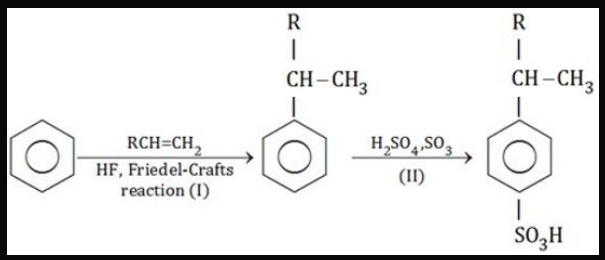
Bromination of phenol is solvent dependent reaction i.e. polarity of solvents.

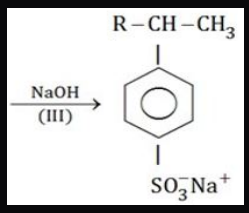


Q.36 Answer:

I, II, III

Solution:



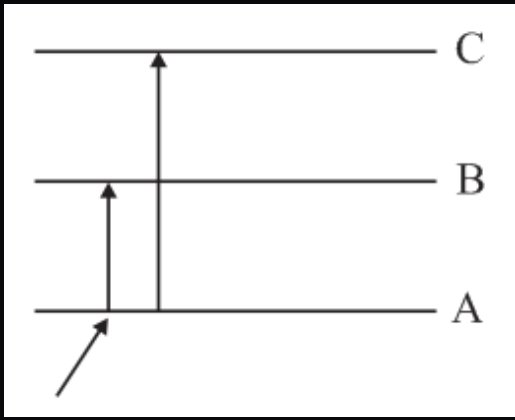


Q.37 Answer:

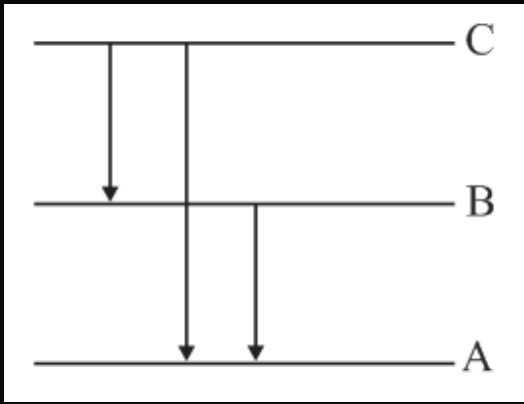
Statement- 1 is True, Statement-2 is True, Statement-2 is a correct explanation for Statement -1

Solution:

Absorption transition



Two possibilities in absorption transition.



Three possibilities in emission transition.

Therefore, absorption transition < emission.

Q.38 Answer:

I and IV

Solution:

In I and IV, amine is not formed.

I. Hydrolysis of RCN; $\mathrm{RNC} \rightarrow \mathrm{RCOOH} + \mathrm{NH_3}$

II. Reduction of $\mathrm{RCH=NOH}$

$\mathrm{RCH=NOH} \rightarrow \mathrm{RCH_2\,NH_2} + \mathrm{H_2O}$

III. Hydrolysis of RNC

$\mathrm{RNC} + 2\mathrm{H_2O} \xrightarrow{\Delta} \mathrm{RNH_2} + \mathrm{HCOOH}$

IV. Hydrolysis of $\mathrm{RCONH_2}$

$\mathrm{RCONH_2} \rightarrow \mathrm{RCOOH} + \mathrm{NH_3}$

Q.39 Answer:

twice the rate of consumption of A.

Solution:

$A + 3B \rightarrow 2C$

Rate of reaction $= -\frac{d(A)}{dt} = -\frac{1}{3}\frac{d(B)}{dt} = +\frac{1}{2}\frac{d(C)}{dt}$

$\therefore \frac{d(C)}{dt} = -\frac{2d(A)}{dt}$

Q.40 Answer:

$\frac{K_2}{K_1}$

Solution:

(i) $XeF_6(g) + H_2O(g) \rightleftharpoons XeOF_4(g) + 2HF(g); K_1$

(ii) $XeO_4(g) + XeF_6(g) \rightleftharpoons XeOF_4(g) + XeO_3F_2; K_2$

Required equation $XeO_4(g) + 2HF(g) \rightleftharpoons XeO_3F_2(g) + H_2O(g)$

It is obtained by subtracting (i) from (ii), hence equilibrium constant of the required equation will be $\frac{K_2}{K_1}$

Q.41 Answer:

smaller radius of fluorine, high electron density

Solution:

Fluorine has low EA than chlorine because of smaller size of fluorine and compact 2p-orbital where interelectronic repulsion is more

Q.42 Answer:

24 u

Solution:

Mol. wt. of $CH_4 = 16$

Mol. wt. of $C_2H_4 = 28$

$\therefore 20 = \frac{16x+28y}{x+y}$

or $16x + 28y = 20x + 20y$

or $4x = 8y$

or $x = 2y$

In the gaseous mixture when the mole ratio of CH_4 **and** C_2H_4 **is** $y : x$

then avg. mol. wt $= \frac{16y+28x}{x+y} = \frac{16y+56y}{3y}$

$= \frac{72y}{3y} = 24$ u

Q.43 Answer:

$8.5 \times 10^{-5}M$

Solution:

$K_{sp}(SrCrO_4) = [Sr^{2+}][CrO_4^{2-}]$

$[CrO_4^{2-}] = \frac{7\times10^{-6}}{0.1} = 7 \times 10^{-5}$

$K_{sp}(BaCrO_4) = [Ba^{2+}][CrO_4^{2-}]$

$[CrO_4^{2-}]_{total} \approx [CrO_4^{2-}]$ **from** $SrCrO_4$

$[Ba^{2+}] = \frac{K_{sp}(BaCrO_4)}{[CrO_4^{2-}]}$

$[Ba^{2+}] = \frac{6\times10^{-9}}{7\times0^{-5}} = 0.85 \times 10^{-4}$
$= 8.5 \times 10^{-5}M$

Q.44 Answer:

CH_3COCl

Solution:

Acetic acid reacts with PCl_5 to form acetyl chloride.

$CH_3COOH + PCl_5 \longrightarrow CH_3COCl + POCl_3 + HCl$

The -COOH group is converted into -COCl group . Side products are phosphorous oxychloride and HCl. This reaction takes place at a temperature nearly 60 degrees celsius.

Q.45 Answer:

Statement 1 is true, Statement 2 is true; Statement 2 is correct explanation for Statement 1.

Solution:

Secondary amines are more basic in aqueous solution as compared to tertiary amines.

Alkyl group exhibits +I effect.

Due to the increase in the electron density over nitrogen because of the availability of lone pair for donation.

In aqueous solutions of tertiary amines the bulkier alkyl groups result in increased strain, decreasing the availability of lone pair inturn decreasing the basicity.

Q.46 Answer:

—

Solution:

Praseodymium (Pr), erbium (Er) and gadolinium (Gd) are lanthanoids.

Q.47 Answer:

—

Solution:

In the aqueous solution

	AB	$\rightleftharpoons$	A^+	$+$	B^-
$(t = 0)$	1		0		0
$(t = \text{any time})$	$1-\alpha$		α		α

Now,

$i = (1-\alpha) + \alpha + \alpha$

$\therefore i = 1 + \alpha \quad \{\alpha = \text{degree of ionisation}\}$

As we know,

Mole fraction of the solute = $(X_a) = \frac{i\times n_a}{i\times n_a + n_b} = \frac{i}{i+3}$

Relative lowering of vapour pressure = Mole fraction of the solute.

$0.4 = \frac{i}{i+3}$

$\alpha = 2 - 1 = 1$

% Ionisation $= 100\%$

Q.48 Answer:

—

Solution:

Suppose a substance

$\mathrm{A} \xrightarrow{\mathrm{T}}$ product.

According to the rate law:

Rate of reaction, $\mathrm{r} = \mathrm{k}\,[\mathrm{A}]^{\mathrm{n}} \ \ldots . (1)$

Assume n **is order of the reaction. When the concentration of reactant** A **is increased by** 1.5 **times, then the rate increases by** 2.25 **times.**

The rate law for this condition, $2.25\ \mathrm{r} = \mathrm{k}\,[1.5\ \mathrm{A}]^{\mathrm{n}} \ \ldots . (2)$.

Dividing the equation (2) **by equation** (1)

$\frac{2.25\ \mathrm{r}}{\mathrm{r}} = \frac{\mathrm{k}\,[1.5\ \mathrm{A}]^{\mathrm{n}}}{\mathrm{k}\,[\mathrm{A}]^{\mathrm{n}}}$

$\Rightarrow 2.25 = [1.5]^{\mathrm{n}}$

$\Rightarrow \mathrm{n} = 2.$

Q.49 Answer:

—

Solution:

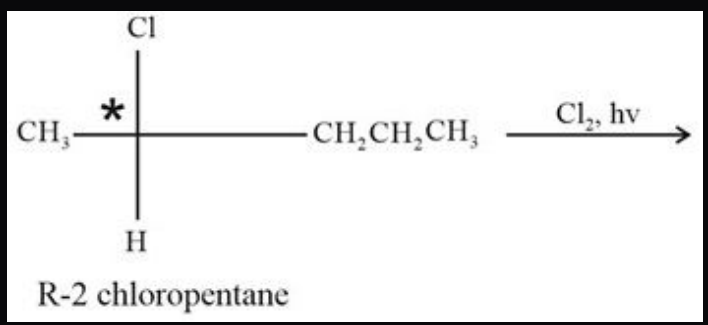
Number of orbital for $n = 3$ **is** $= n^2 = 9$

Number of electrons for $\mathrm{n} = 3$ **and** $\mathrm{m_s} = -\frac{1}{2} = 9$

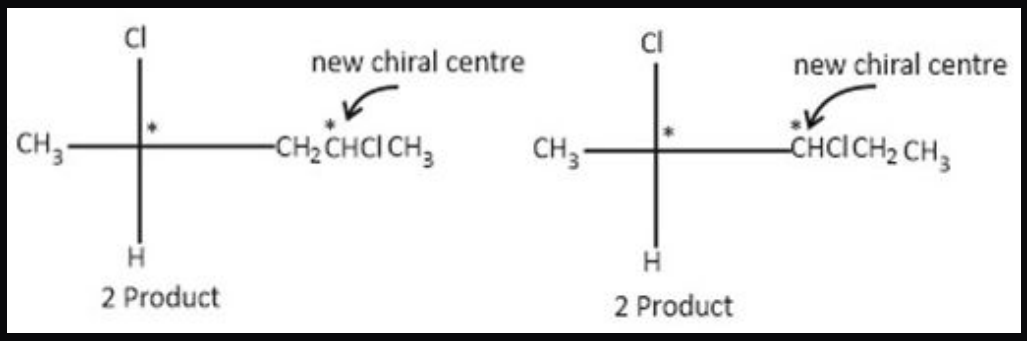
Q.50 Answer:

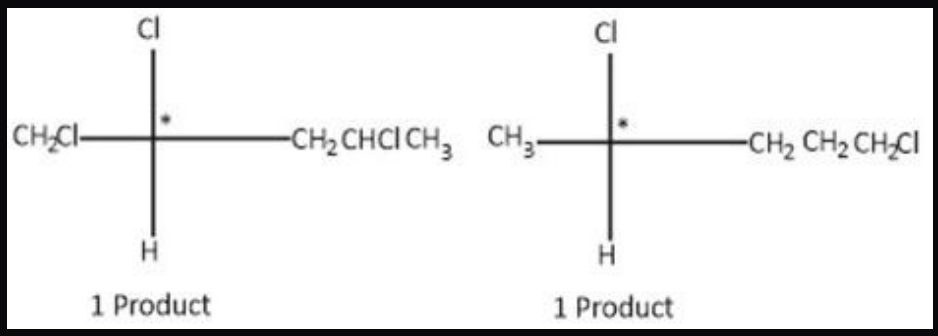
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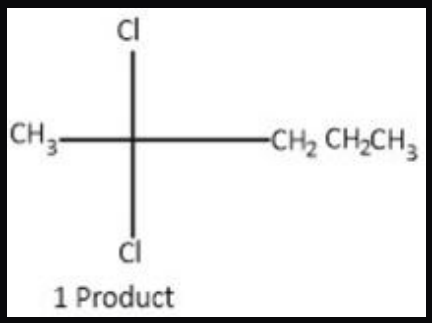
Solution:



R-2 chloropentane







Total 7 products

Q.51 Answer:

$\frac{7^3}{5^3.4^6}$

Solution:

$|AA^T| = |5\text{I}| \Rightarrow |A|^2 = 5^3$ **....(1)**

$3A^{-1} = 2A^T - \text{adj}(4B)$

Multiplying by A on both sides, we get

$3\text{I} = 2AA^T - A\text{adj}(4B)$

$\Rightarrow A^2\text{adj}(4B) = 7\text{I}$

$|A|^2|\text{adj}4B| = 7^3$

$5^3 4^6 |B|^2 = 7^3$ **(using (1))**

$|B|^2 = \frac{7^3}{5^3.4^6}$

Q.52 Answer:

$\sqrt{3}$

Solution:

$\mathrm{I} = \int_{\log\sqrt{\pi/2}}^{\log\sqrt{\pi}} \mathrm{e}^{2\mathrm{x}} \sec^2\left(\frac{1}{3}\mathrm{e}^{2\mathrm{x}}\right) \mathrm{dx}$

Put $\mathrm{e}^{2\mathrm{x}} = \mathrm{t} \Rightarrow 2\mathrm{e}^{2\mathrm{x}}\mathrm{dx} = \mathrm{dt}$

When $\mathrm{x} = \log\sqrt{\pi/2}, \mathrm{t} = \mathrm{e}^{2\log\sqrt{\pi/2}} = \mathrm{e}^{\log\pi/2} = \frac{\pi}{2}$

When $\mathrm{x} = \log\sqrt{\pi}, \mathrm{t} = \mathrm{e}^{2\log\sqrt{\pi}} = \pi$

$\therefore \mathrm{I} = \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2}\sec^2\left(\frac{1}{3}\mathrm{t}\right)\mathrm{dt} = \frac{1}{2}\cdot\frac{1}{\frac{1}{3}}\left[\tan\frac{\mathrm{t}}{3}\cdot\right]_{\pi/2}^{\pi}$

$= \frac{3}{2}\left[\tan\frac{\pi}{3} - \tan\frac{\pi}{6}\right] = \frac{3}{2}\left[\sqrt{3} - \frac{1}{\sqrt{3}}\right] = \sqrt{3}$

Q.53 Answer:

$\frac{x}{3} + \frac{1}{3}$

Solution:

Since, $[m+n] = [m] + n,$ **where** n **is an integer**

and $[p] = p - \{p\}$

$\Rightarrow \left[1^2x + 1^2\right] = 1^2x + 1^2 - \left\{1^2x\right\}$

$\left[2^2x + 2^2\right] = 2^2x + 2^2 - \left\{2^2x\right\}$

$\left[n^2x + n^2\right] = n^2x + n^2 - \left\{n^2x\right\}$

Therefore,

$\lim\limits_{n\to\infty} \frac{\left[1^2x+1^2\right]+\left[2^2x+2^2\right]+....+\left[n^2x+n^2\right]}{n^3}$

$= \lim\limits_{n\to\infty} \frac{\sum_{r=1}^{n}\left(r^2x+r^2\right) - \sum_{r=1}^{n}\left\{r^2x\right\}}{n^3}$

Since fractional part of any number lies between 0 **and** $1.$

Therefore, $\sum_{r=1}^{n}\left\{r^2x\right\} < n \Rightarrow \lim\limits_{n\to\infty} \frac{\sum_{r=1}^{n}\left\{r^2x\right\}}{n^3} = 0$

$= \lim\limits_{n\to\infty} \frac{\sum_{r=1}^{n}\left(r^2x+r^2\right)}{n^3}$

$= \lim\limits_{n\to\infty} \frac{(x+1)}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} = \frac{x+1}{3}$

Q.54 Answer:

-19

Solution:

$(1-x)^8(1+x)^{12} = \left(1-x^2\right)^8(1+x)^4$

$= \left(1 - 8x^2 + 28x^4 \ldots.\right)\left(1 + 4x + 6x^2 + 4x^3 + x^4\right)$

Coefficient of $x^4 = 1 - 48 + 28 = -19$

Q.55 Answer:

$(2,2,4)$

Solution:

$f(x) = x^3 + ax^2 + bx + c$

$f'(x) = 3x^2 + 2ax + b \geq 0,\ \forall x \in R$

$D \leq 0$

$4a^2 - 12b \leq 0$

$a^2 \leq 3b$

From the given options, $(2,2,4)$ **satisfies the above condition**

Q.56 Answer:

2. $(7!)$

Solution:

Let the arrangement be $x_1x_2x_3x_4x_5x_6x_7x_8$

Clearly, 5 **should occupy the position**x_4 **or** x_5 **. Thus required number of ways are**

$= {}^2C_1 \times 7!$

$= 2(7!)$

Q.57 Answer:

$\mathrm{f(x)} = \frac{\mathrm{x}}{\mathrm{e^x}-1} + \frac{\mathrm{x}}{2} + 1$

Solution:

$$\mathrm{f(x)} = \log\left(\frac{1-\mathrm{x}}{1+\mathrm{x}}\right)$$

$$\therefore \mathrm{f(-x)} = \log\left(\frac{1+\mathrm{x}}{1-\mathrm{x}}\right) = -\log\left(\frac{1-\mathrm{x}}{1+\mathrm{x}}\right) = -\mathrm{f(x)}$$

∴ It is an odd function

(2) $\mathrm{f(x)} = \log\left(\mathrm{x} + \sqrt{\mathrm{x}^2+1}\right) \Rightarrow \mathrm{f(-x)} = \log\left(-\mathrm{x} + \sqrt{\mathrm{x}^2+1}\right)$

$$f(x) + f(-x) = \log\left(x + \sqrt{x^2+1}\right) + \log\left(-x + \sqrt{x^2+1}\right)$$

$$= \log\left\{\left(x + \sqrt{x^2+1}\right)\left(-x + \sqrt{x^2+1}\right\} = \log\left(x^2 + 1 - x^2\right)\right.$$

$$= \log(1) = 0 \quad \therefore \mathrm{f(-x)} = -\mathrm{f(x)}$$

∴ It is an odd function

(3) $\mathrm{f(x)} = \frac{\mathrm{x}}{\mathrm{e^x}-1} + \frac{\mathrm{x}}{2} + 1;$

$$f(-x) = \frac{-x}{e^{-x}-1} - \frac{x}{2} + 1 = \frac{-xe^x}{1-e^x} - \frac{x}{2} + 1$$

$$\mathrm{f(x)} - \mathrm{f(-x)} = \left\{\frac{\mathrm{x}}{\mathrm{e^x}-1} + \frac{\mathrm{x}}{2} + 1\right\} - \left\{\frac{-\mathrm{xe^x}}{1-\mathrm{e^x}} - \frac{\mathrm{x}}{2} + 1\right\}$$

$$\frac{x\left(1-e^x\right)}{e^x-1} + x = -x + x = 0$$

$$\therefore \mathrm{f(-x)} = \mathrm{f(x)}$$

Hence, it is an even function

(4) $\mathrm{f(x)} = \mathrm{e^{2x}} + \sin\mathrm{x}$

$$f(-x) = e^{-2x} + \sin(-x) = e^{-2x} - \sin x \neq \pm f(x)$$

Q.58 Answer:

6

Solution:

Let the first term be a and the common difference be d, then

$20 = \frac{1}{a+4d}$ and $50 = \frac{1}{a+5d}$

On solving, we get,

$d = -\frac{3}{100}$ and $a = \frac{17}{100}$

Hence, $t_r = \frac{100}{20-3r} > 0 \Rightarrow r < \frac{20}{3} \Rightarrow r_{\max} = 6$

So, sum is maximum for $n = 6$, as after this terms are negative

Q.59 Answer:

$\left(2 - 2\sqrt{2}\right)$ units

Solution:

Let α be an acute angle that the given line make with X-axis. Then, $\cos^2\alpha + \cos^2 120^\circ + \cos^2 135^\circ = 1$

$\Rightarrow \cos^2\alpha = \pm\frac{1}{2}$ but α is acute angle

$\therefore \cos\alpha = +ve$

$\cos\alpha = \frac{1}{2} \Rightarrow \alpha = 60^\circ$

So, direction of cosine of the given line are $\cos 60^\circ, \cos 120^\circ, \cos 135^\circ$

i.e., $\frac{1}{2}, -\frac{1}{2}, \frac{-1}{\sqrt{2}}$

Now, the projection of AB on the line $= \frac{1}{2}(4+1) - \frac{1}{2}(3-2) - \frac{1}{\sqrt{2}}(5-1)$

$= \frac{5}{2} - \frac{1}{2} - 2\sqrt{2} = \left(2 - 2\sqrt{2}\right)$ units

Q.60 Answer:

$x - y = 0$

Solution:

Line Joining $(1, 2)$ and $(2, 1)$ and line Joining $(3, 4)$ and $(4, 3)$ are parallel.

Also, the perpendicular bisector of any two parallel chords is axis of parabola.

Hence, axis of the parabola is $x - y = 0$

Q.61 Answer:

6

Solution:

$\vec{r} \cdot \hat{i} = 2\vec{r} \cdot \hat{j} = 4\vec{r} \cdot \hat{k}$

Let $\vec{r} = x\hat{i} + y\hat{j} + z\vec{k}$

$x = 2y = 4z$

$\left|\vec{r}\right| = \sqrt{84} \Rightarrow \sqrt{(4z)^2 + (2z)^2 + z^2} = \sqrt{84}$

$z = 2; y = 4; x = 8$

$\vec{r} = 8\hat{i} + 4\hat{j} + 2\hat{k}$

$\vec{r} \cdot \left(2\hat{i} - 3\hat{j} + \hat{k}\right) = 16 - 12 + 2 = 6$

Q.62 Answer:

$> \frac{1}{4}$

Solution:

A_1 : **white ball is drawn from bag**

A_2 : **Red ball is drawn from bag** 1

B : **2 balls are drawn from bag 2 and found to be red.**

$P(B) = P(A_1) \cdot P(B/A_1) + P(A_2)P(B/A_2)$

$$\Rightarrow P(A_1/B) = \frac{\frac{3}{5} \cdot \frac{{}^5C_2}{{}^{10}C_2}}{\frac{3}{5} \cdot \frac{{}^5C_2}{{}^{10}C_2} + \frac{2}{5} \cdot \frac{{}^6C_2}{{}^{10}C_2}} = \frac{1}{2}$$

Q.63 Answer:

$\frac{17}{27}$

Solution:

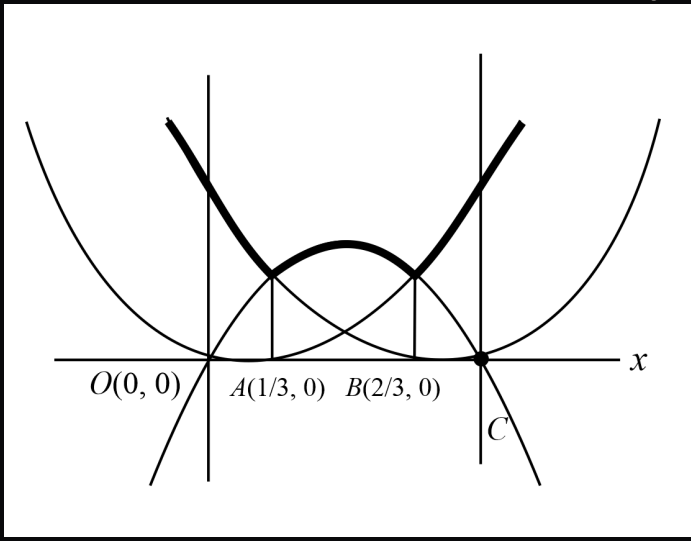
$f(x) = \textbf{Max}\left\{x^2, (1-x)^2, 2x(1-x)\right\}$

coordinate of A, $(1-x)^2 = 2x(1-x)$

$(1-x)(1-x-2x) = 0$

$x = \frac{1}{3},\ A\left(\frac{1}{3}, 0\right)$ **coordinate of** B

$2x(1-x) = x^2, 2(1-x) = x, 2-3x = 0,\ x = \frac{2}{3}$



$B\left(\frac{2}{3}, 0\right)$ Req. Area $= \int_0^{1/3} (1-x)^2 dx + \int_{1/3}^{2/3} 2x(1-x)dx + \int_{\frac{2}{3}}^{1} x^2 dx$

$= \frac{17}{27}$

Q.64 Answer:

$-\frac{1}{14}$

Solution:

We have, $8f(x) + 6f\left(\frac{1}{x}\right)$**=**$x$**+5 for all** x **...(i)**

Therefore, $8f\left(\frac{1}{x}\right) + 6f(x) = \frac{1}{x}$**+5 ...(ii)**

From Eqs. (i) and (ii), we have

$f(x) = \frac{1}{28}\left(8x - \frac{6}{x} + 10\right)$

Now, $y = x^2\ f(x)$

$\Rightarrow \quad y = \frac{1}{28}\left(8x^3 - 6x + 10x^2\right)$

$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{28}\left(24x^2 + 20x - 6\right)$

$\therefore \quad \left(\frac{dy}{dx}\right)_{x=-1} = \frac{1}{28}\left(24 - 20 - 6\right)$

$= -\frac{1}{14}$

Q.65 Answer:

$x^2 + y^2 = 4a^2$

Solution:

We know that centroid divides the median in the ratio $2:1$.

Radius of the circle $= \frac{2}{3}\times$ length of median $= \frac{2}{3} \times 3a = 2a$

Centre of the (given) circle is $C(0,0)$. Therefore the equation of the circle $(x-0)^2 + (y-0)^2 = (2a)^2 \Rightarrow x^2 + y^2 = 4a^2$

Q.66 Answer:

1

Solution:

We can write Δ as,

$$\Delta = \sin^2\alpha \sin^2\beta \sin^2\gamma \begin{vmatrix} 1 & \cot\alpha & \cot^2\alpha \\ 1 & \cot\beta & \cot^2\beta \\ 1 & \cot\gamma & \cot^2\gamma \end{vmatrix}$$

$= \sin^2\alpha \sin^2\beta \sin^2\gamma (\cot\beta - \cot\alpha)(\cot\gamma - \cot\alpha)(\cot\gamma - \cot\beta)$

$= \sin(\alpha-\beta)\sin(\alpha-\gamma)\sin(\beta-\gamma)$

It is clear from here that Δ cannot exceed 1.

$[\because \sin\theta \times 1$, for any $\theta \in \mathbf{R}]$

Q.67 Answer:

none of these

Solution:

$\because \ (x,x) \notin R$

$\therefore$ **not reflexive**

If $2x - y = 10 \Rightarrow 2y - x = 10$

$\therefore$ **not symmetric**

If $2x - y = 10$ **and** $2y - z = 10$

$\Rightarrow \quad 2x - z = 10 \quad \therefore$ **not transitive**

Q.68 Answer:

$\left(\frac{3}{2}, \frac{1}{2}, -1\right)$

Solution:

We have,

$\int \frac{2\cos x-\sin x+\lambda}{\cos x+\sin x-2}dx = A\log_e(|\cos x+\sin x-2|)+Bx+C$

Differentiating R. H. S. w.r.t. x, we get

$\frac{d}{dx}[A\log_e(|\cos x+\sin x-2|)+Bx+C] = A\frac{\cos x-\sin x}{\cos x+\sin x-2}+B$

$= \frac{A\cos x-A\sin x+B\cos x+B\sin x-2B}{\cos x+\sin x-2}$

$\therefore \frac{2\cos x-\sin x+\lambda}{\cos x+\sin x-2} = \frac{A\cos x-A\sin x+B\cos x+B\sin x-2B}{\cos x+\sin x-2}$

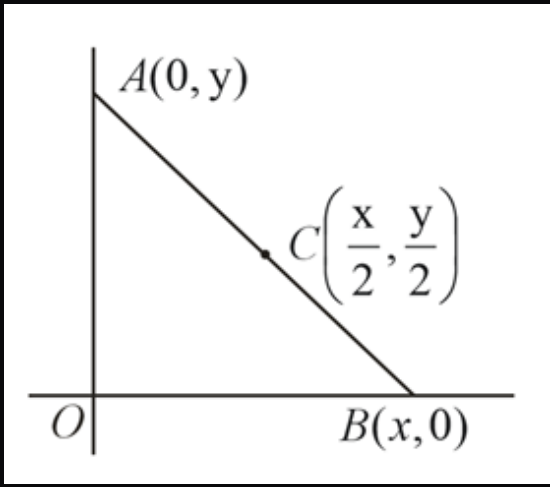
So, $A+B=2,\ B-A=-1,\ \lambda=-2B$

$\Rightarrow A=\frac{3}{2},\ B=\frac{1}{2},\ \lambda=-1$

Q.69 Answer:

$\frac{b}{\sqrt{2}}\frac{\text{cm}}{\text{s}}$

Solution:



Let the stick be $AB = a$ cm.

Coordinates of mid-point of AB is $D\left(\frac{x}{2},\frac{y}{2}\right)$.

Velocity of point B is $b\frac{\text{cm}}{\text{s}}$. Therefore,

$\frac{dx}{dt} = b$

Now, in ΔAOB

$AB^2 = OA^2+OB^2$

$\Rightarrow a^2 = x^2+y^2$

Differentiating both sides, we have

$2x\frac{dx}{dt}+2y\frac{dy}{dt}=0$

$\Rightarrow x\frac{dx}{dt}+y\frac{dy}{dt}=0$

$\Rightarrow bx+y\frac{dy}{dt}=0$

When the stick in equally inclined then $x=y$

$by+y\frac{dy}{dt}=0$

$\Rightarrow \frac{dy}{dt}=-b$

Now coordinates of mid-point of AB is $D\left(\frac{x}{2},\frac{y}{2}\right)$. Therefore, velocity vector at D is

$\frac{1}{2}\left(\frac{dx}{dt}\right)+\frac{1}{2}\left(\frac{dy}{dt}\right)$.

Resultant velocity of mid point of AB is

$\sqrt{\left(\frac{1}{2}\cdot\frac{dx}{dt}\right)^2+\left(\frac{1}{2}\cdot\frac{dy}{dt}\right)^2}$

$= \sqrt{\frac{b^2}{4}+\frac{b^2}{4}}$

$= \frac{b}{\sqrt{2}} \frac{\text{cm}}{\text{s}}$

Q.70 Answer:

$\tan y \tan x = c$

Solution:

Given, $\frac{\sec^2 x}{\tan x} dx = -\frac{\sec^2 y}{\tan y} dy$

$\Rightarrow \int \frac{\sec^2 x}{\tan x} dx = -\int \frac{\sec^2 y}{\tan y} dy$

Put $\tan x = u$

$\Rightarrow \sec^2 x \, dx = du$

And $\tan y = v$

$\Rightarrow \sec^2 y \, dy = dv$

$\therefore \int \frac{du}{u} = -\int \frac{dv}{v}$

$\Rightarrow \log u = -\log v + \log c \Rightarrow uv = c$

$\therefore \tan x . \tan y = c$

Q.71 Answer:

—

Solution:

Any natural number is either of the form 3k or 3k - 1 or 3k + 1.

Sum of two numbers will be divisible by 3 if and only if, either both are of the form 3k ($^4C_2$ ways).

OR

One is of the form 3k - 1 and other is of the form 3k + 1($^4C_1$ x $^4C_1$ ways).

Hence, total number of required ways

$= {}^4\text{C}_2 + {}^4\text{C}_1 \times {}^4\text{C}_1 = 6 + 16 = 22$

$\Rightarrow \quad \text{k} = 22$

$\therefore \ \text{k} - 16 = 6$

Q.72 Answer:

—

Solution:

Since the triangle is an equilateral triangle, so ortho-center, circum-center, in-center, and centroid all coincide at a single point.

Thus, $\text{Centroid} = \left(\frac{\vec{a}+\vec{b}+\vec{c}}{3}\right) = \text{Orthocentre} = \vec{0}$

$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$

Q.73 Answer:

—

Solution:

$n(A) = 5,\ n(B) = 4$

So divide 5 **elements in** 4 **group** $= 1, 1, 1, 2$

Total number of ways $= \left(\frac{5!}{(11)^3 2!.3!}\right) 4!$

$= 240$

$\lambda = 240$

Sum of digits $= 2 + 4 + 0 = 6$

Q.74 Answer:

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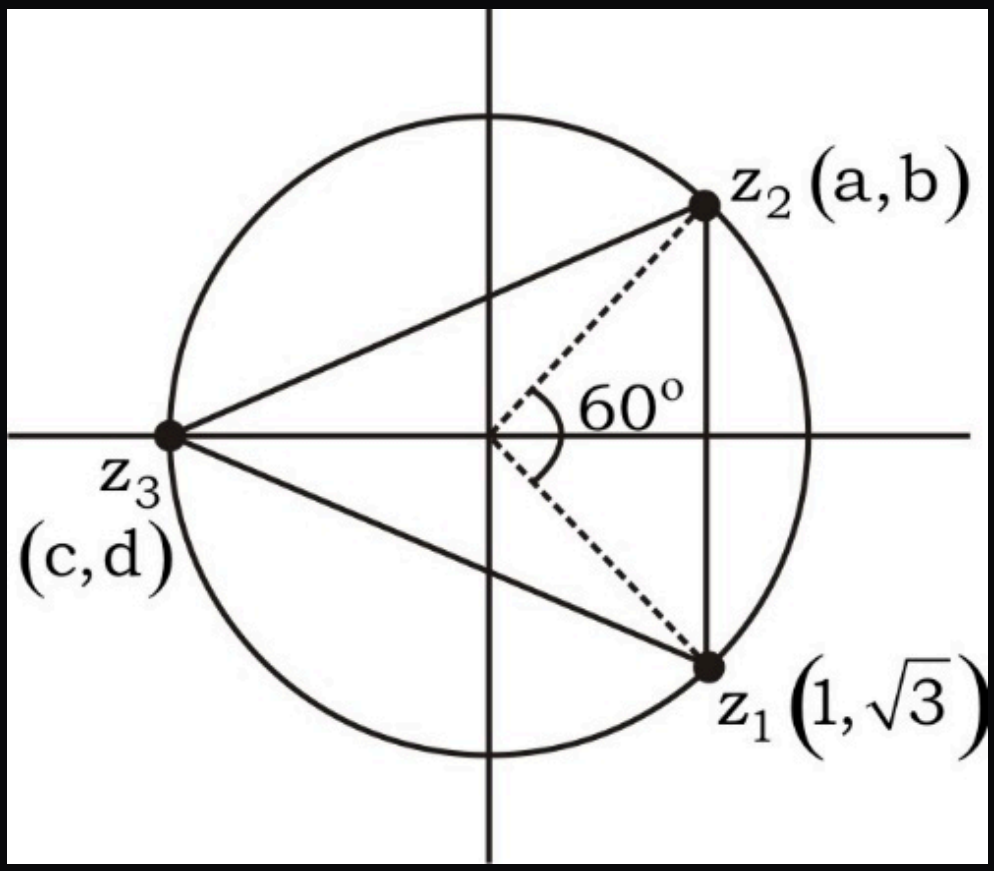
Solution:

$\therefore \quad z_2 = i + i\sqrt{3}$

$z_3 = -2$

$a = 1,\ b = \sqrt{3}$

$c = -2,\ d = 0$



Q.75 Answer:

—

Solution:

$x + \frac{1}{x} = 2 \Rightarrow x = 1 \ldots . \left[x + \frac{1}{x} \geq 2 \text{ as equality occurs only when } x = 1\right]$

$\Rightarrow \sin^{-1} x = \frac{\pi}{2}$

Similarly, $y = -1 \ldots . \left[y + \frac{1}{y} \leq -2 \text{ as equality occurs only when } y = -1\right]$

$\Rightarrow \cos^{-1} y = \pi$

$\sin^{-1} x + \cos^{-1} y = \frac{\pi}{2} + \pi$

$= \frac{3\pi}{2}$

$\Rightarrow m = \frac{3}{2} = 1.5$